# A Latent Oscillator Measurement Model to Simulate Emotional-Expression Score Dynamics in Video

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**Abstract**

Facial-expression classifiers convert video of a person into multivariate time series of classifier scores. These scores contain measurement error from the classifier, the video, and the recording conditions, and the latent structure that generated them is unknown. Empirical score series alone therefore cannot show whether the score channels are indicators of a smaller number of latent expressive processes, or whether an analysis method would recover those processes if they existed. We introduce the Latent Oscillator Measurement Model (LOMM), a data-generating model that specifies three components separately: latent dynamics as damped, undamped, or amplifying linear oscillators; a time-varying activity level that determines how strongly each latent dimension is expressed; and a factor-analytic observation model with fallible indicators. Study 1 used four-fold cross-fitting with classifier scores from 100 MAFW videos to calibrate LOMM and compare generated series with MAFW series excluded from calibration. Median series-level plausibility and coverage were 0.970 and 0.920 for LOMM, compared with 0.510 and 0.370 for a static logistic-normal generator. Study 2 generated continuous indicators with a known number of dimensions and known indicator assignments and asked whether four methods, Dynamic Exploratory Graph Analysis (DynEGA), static EGA, GraphicalVAR, and GIMME, recovered that structure through a common community-detection step. At 100 observations per clip, with failed or timed-out fits counted as incorrect, overall correct-dimension recovery was 0.939 for DynEGA, 0.884 for static EGA, 0.777 for GraphicalVAR, and 0.176 for GIMME. Replacing a shared initial state with independent initial states reduced recovery for DynEGA, static EGA, and GraphicalVAR, so a shared initial state overstates recovery. LOMM lets researchers test whether a method recovers a known structure before empirical score patterns are interpreted psychologically.

# Introduction

Facial-expression classifiers convert video of a person into multivariate time series. Each frame yields a vector of scores, one for each label in a chosen label set, at a temporal resolution that manual coding cannot match.

Earlier classifiers returned scores only for the categories in their training data. Vision-language models such as CLIP can score any label supplied as text (Foteinopoulou & Patras, 2023; Li et al., 2023; Radford et al., 2021). The scores are nonetheless measurements: they depend on the model, the label set, the video-processing pipeline, image quality, and recording conditions, and their accuracy varies with model adaptation, label choice, and evaluation data (Foteinopoulou & Patras, 2023; Li et al., 2023). A pattern among score channels may therefore reflect expressive behavior, the measurement process, or both. In this paper, classifier scores are treated as fallible measurements of visible expression, not as measures of subjective emotional experience.

This raises a psychometric question. Score channels can be analyzed as separate variables, or they can be treated as fallible indicators of a smaller number of latent expressive processes (Mehu & Scherer, 2015; Nesselroade et al., 2002; Russell, 2003). Measurement error changes the structure that dynamic methods recover (Kankaanpää et al., 2026). Empirical data cannot settle the question, because the true number of latent dimensions and the true assignment of indicators to dimensions are unknown, so there is no way to check whether a method recovered them.

We introduce the Latent Oscillator Measurement Model (LOMM) to make that check possible. LOMM is a data-generating model for multivariate intensive longitudinal data with an explicit measurement model. It simulates latent processes that change over time, specifies how strongly each process is expressed at each observation, and maps the processes onto fallible observed indicators. The number of dimensions, the indicator assignments, the dynamics, the activity pattern, the loadings, and the measurement error are all fixed before a recovery method is applied. LOMM combines the damped linear oscillator model of intraindividual dynamics (Boker, 2015; Boker & Nesselroade, 2002) with a factor-analytic observation model (Nesselroade et al., 2002; Zhang et al., 2008).

We use LOMM in two studies (Figure 1). In Study 1 the MAFW classifier-score series are the target (Liu et al., 2022). Each of four folds uses 75 MAFW videos to calibrate LOMM and 25 videos to evaluate the generated series. A static logistic-normal generator provides a comparison without temporal dependence. Eleven statistics computed on scores and log-ratio coordinates define series-level plausibility and coverage. In Study 2 the LOMM design is the target. LOMM generates continuous indicators with a known number of dimensions and a known assignment of indicators to dimensions, and four methods estimate a network

among the indicators and group them into communities: Dynamic Exploratory Graph Analysis (DynEGA), static EGA, GraphicalVAR, and GIMME. The question is whether the recovered communities match the structure that generated the data. Static EGA ignores temporal order and is computationally inexpensive; it shows how much structure is recoverable from contemporaneous indicator levels alone. Study 2 also examines two conditions that limit the main result: whether a shared initial state across dimensions inflates recovery, and whether the indicator grouping is preserved after the indicators are transformed to bounded scores that sum to one, as classifier scores do. Because the MAFW videos contain a median of 35 scored observations, Study 2 also reports a secondary analysis with 25 observations per clip.

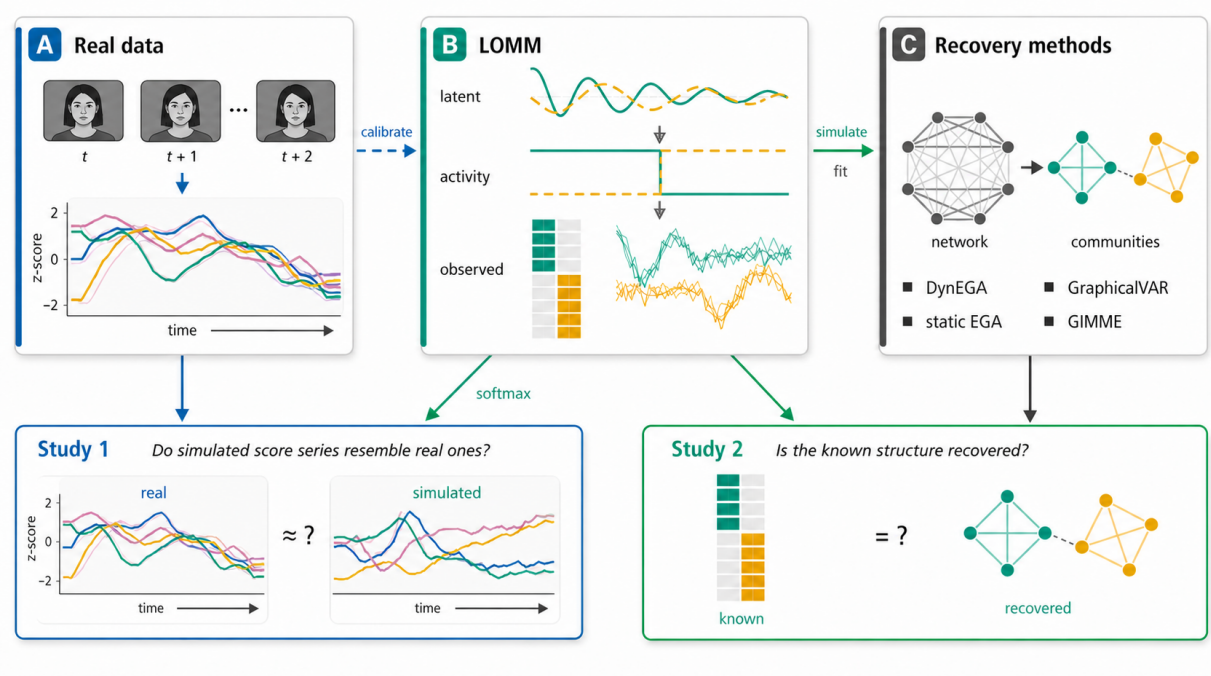



Figure 1: Overview of the two studies. (A) Real data: a classifier scores each sampled frame of an MAFW video, giving one score series per label. (B) LOMM: latent oscillators, an activity schedule that sets how strongly each is expressed at each observation, and observed indicators with a known assignment to dimensions (colored blocks). (C) Recovery methods: DynEGA, static EGA, GraphicalVAR, and GIMME each estimate a network among the indicators and group them into communities. Study 1 uses MAFW calibration videos to set the generator and compares simulated score series with MAFW evaluation series. Study 2 fits the four methods to simulated indicators and compares the recovered communities with the known assignment.

# The Latent Oscillator Measurement Model

LOMM represents a standard psychometric situation: several observed variables are fallible indicators of a smaller number of latent processes, and those processes change over time. As an illustration, suppose a simulation has three latent dimensions labeled approach-affiliation, threat-defense, and orienting-novelty,

and the anger, fear, disgust, and anxiety channels are assigned as indicators of threat-defense. Those four channels would covary over time because they load on the same latent dimension, and each would also carry channel-specific measurement error. A recovery method should group the four channels together and should find three dimensions. These labels only illustrate a possible simulation design; whether MAFW scores have this structure is not tested here.

LOMM generates data in three layers (Figure 2). The latent layer describes how each latent dimension changes over time. The activity layer determines how strongly that dimension is expressed at each observation. The observation layer creates multiple indicators with prespecified loadings and measurement error. Keeping these layers separate allows a researcher to change the dynamics, the expression pattern, or the measurement quality while holding the other components fixed.

*Notation.* Clips are indexed by $i = 1, \dots, N$ and observations within a clip by $n = 1, \dots, T$, at times $t_n = (n-1)\Delta$, where $\Delta$ is the observation interval. Latent dimensions are indexed by $f = 1, \dots, m$. Continuous indicators are indexed by $j = 1, \dots, p$; in Study 2, $p = mk$ with $k$ indicators per dimension. Bounded scores have $J$ channels. Bold symbols are vectors or matrices.

For clip $i$ and latent dimension $f$, let $x_{if}(t)$ denote the latent level and $\dot{x}_{if}(t)$ its rate of change. LOMM treats each dimension as a linear second-order oscillator:

$$\ddot{x}_{if}(t) = \eta_f x_{if}(t) + \zeta_f \dot{x}_{if}(t) + q_{if,n}, \qquad t \in [t_n, t_{n+1}). \tag{1}$$

Equation 1 is the damped linear oscillator used in research on intraindividual dynamics (Boker & Nesselroade, 2002). The position-feedback parameter $\eta_f$ determines the strength of the restoring force toward equilibrium; more negative values produce faster oscillation. The velocity-feedback parameter $\zeta_f$ determines damping: negative values damp the motion, zero leaves it undamped, and positive values amplify it. The term $q_{if,n}$ is a random acceleration input, drawn independently for each interval between observations $n$ and $n+1$ and held constant within that interval; it produces irregular peaks and reversals in the trajectory. When $\eta_f$ and $\zeta_f$ are both negative, the trajectory returns toward equilibrium, with oscillation when the characteristic roots are complex. In the conditions tested here, positive $\zeta_f$ produces amplitudes that grow over the observed record; we call this finite-horizon amplification because the record ends before the growth becomes extreme. Appendix A gives the exact discrete-time transition used to advance the system between observations, the stability conditions, and the initialization rules.

The activity layer multiplies the latent level at each observation by a nonnegative activity value, $\tilde{x}_{if,n} =$

$a_{if,n}x_{if,n}$. A latent dimension can therefore remain strongly expressed throughout a clip, remain near a low background level, or alternate with other dimensions. This layer matters for recovery because an abrupt change in activity produces a large change in the observed indicators even when the oscillator itself has not accelerated.

The observation layer is a standard factor model:

$$\mathbf{z}_{in} = \Lambda_i \tilde{\mathbf{x}}_{in} + \varepsilon_{in}, \qquad \varepsilon_{in} \sim \mathcal{N}(\mathbf{0}, \Sigma_i). \tag{2}$$

Row $j$ of $\Lambda_i$ contains the loadings of indicator $j$ on the $m$ latent dimensions, and $\varepsilon_{in}$ is measurement error with covariance $\Sigma_i$. Appendix A gives the complete observation equation, including optional clip-specific baseline and scale terms.

Figure 2 shows the three layers for one simulated clip. In panel A the dimensions differ in their feedback parameters and start from different states: dimension 1 rises to an early peak and decays smoothly, dimension 2 starts in a trough and rebounds, and dimension 3 oscillates faster and is driven mainly by its acceleration inputs. Panel B shows the activity schedule, in which the dimensions take turns being active. Panel C shows the product of the two layers. A latent peak that falls inside an active interval is expressed at full size; the same peak outside the interval is reduced to 8% of its size and is barely visible in the indicators. The steps at the boundaries of the active intervals appear in the indicators although no oscillator has accelerated there. Panel D shows the twelve indicators; each loads mainly on its own dimension, and nine also carry a smaller loading on another dimension. Panel E shows the correlation matrix among the twelve indicators from this clip. The three diagonal blocks reflect the indicator assignments, which is the structure that Study 2 asks the methods to recover. The weaker positive and negative correlations between blocks come from the secondary loadings, so the block structure is not clean; the secondary loadings in this example are larger than in the Study 2 cross-loading condition, which limits them to $\pm 0.10$.

LOMM can output two types of observed variables. The first is the vector of continuous indicators $\mathbf{z}_{in}$, used in the primary Study 2 designs; there, the factor structure that generated the data is defined directly on the analyzed variables. The second is a vector of $J$ positive scores that sum to one, which matches the form of classifier output and is used in Study 1. To produce bounded scores, LOMM maps the latent dimensions to a $(J-1)$-dimensional vector of log-ratio coordinates and applies the softmax function. Because scores that sum to one are compositional data, they are analyzed both as scores and in log-ratio coordinates, which remove the sum constraint. A separate Study 2 condition applies the softmax directly to the $p$ continuous indicators and tests whether their original grouping is preserved. Appendix A gives the softmax, log-ratio,

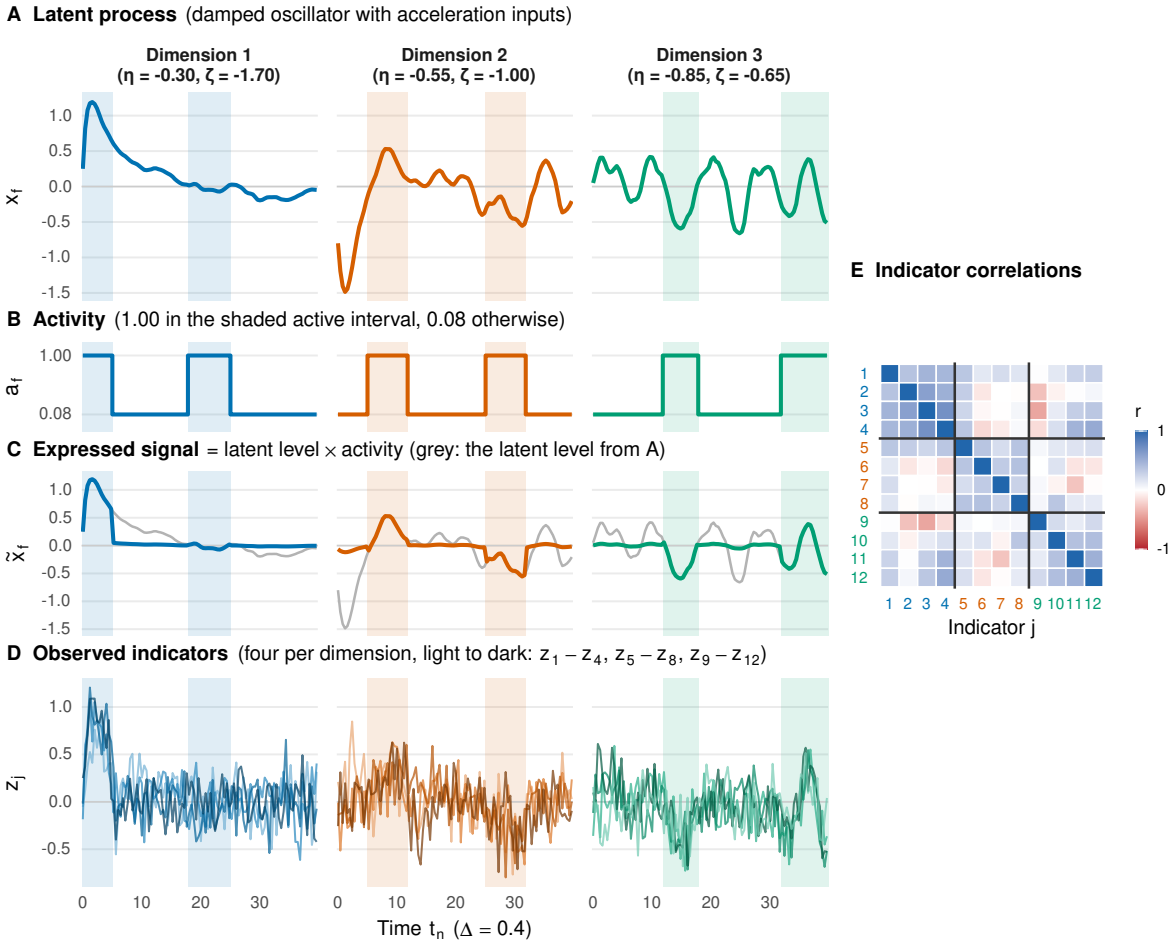



Figure 2: One simulated LOMM clip with three latent dimensions, $T = 100$ observations, and $\Delta = 0.4$. Each column follows one dimension from the latent layer to the observed indicators. (A) Latent trajectory; the oscillator parameters are given in the column headers, and each dimension has its own initial state and acceleration inputs. (B) Activity, 1.00 in the shaded active interval and 0.08 otherwise. (C) Expressed signal, the product of the latent level and the activity value, drawn over the latent level in grey. (D) The four observed indicators of each dimension in shades from light to dark, with primary loadings 0.65 to 0.85, secondary loadings of $\pm 0.30$ or $\pm 0.35$ on another dimension for nine of the twelve indicators, and residual SD 0.20. (E) Correlations among the twelve indicators from this clip, ordered by dimension.

and identification details.

In Study 1, MAFW calibration videos determine the LOMM observation-model geometry and generator settings. The primary Study 2 design sets between-clip baselines to zero and scales to one but lets loadings and residual variances vary slightly across clips. Estimating oscillator, activity, and measurement parameters from empirical data would require the sign, scale, temperature, and frequency constraints stated in Appendix A.

# Study 1: Comparing Simulated and Real Score Series

Study 1 examined whether LOMM, after calibration to MAFW, could generate classifier-score series with statistical properties observed in MAFW videos excluded from calibration. The analysis used 100 videos from MAFW, a large multimodal database of facial-expression video recorded in naturalistic settings (Liu et al., 2022). Frames were scored with the transforEmotion R package (Tomašević et al., 2026) using OpenAI's CLIP ViT-Large/14 model (setting `oai-large`), with up to 50 frames sampled per video. Eleven labels from the MAFW label set served as text prompts: anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, helplessness, and disappointment. For each scored frame, the eleven classifier-score channels sum to one.

Figure 3 shows the classifier-score series for one MAFW video. Frames without a detected face have missing scores, so each video was divided into segments of consecutive scored frames. The analysis contains 3,557 observations in 188 segments. Videos contribute 16 to 50 scored observations, with a median of 35; segment lengths range from 1 to 50 observations, with a median of 13.5. Appendix B gives the face-detection, score-validity, missing-data, and sampling details.

We represented each MAFW and simulated series with a broad panel of statistics covering location, dispersion, short-term change, temporal persistence, temporal irregularity, and cross-channel dimensionality. Eight statistics were computed within channel and video: median level; interquartile range of levels; interquartile range of observation-to-observation changes; lag-1 autocorrelation; and run-length, sample, permutation, and spectral entropy. Their channel-specific values were summarized by the median within each video. Effective rank, robust level variance, and robust change variance described all channels jointly. Each statistic was evaluated on the classifier scores and their isometric log-ratio coordinates, producing 22 features. The main analysis combines the complete feature panel in the plausibility and coverage measures. Appendix C reports the metric-specific comparisons.

We used four-fold cross-fitting. In each fold, 75 MAFW videos formed the calibration set and 25 formed




Figure 3: Classifier-score series from one MAFW video. Panel A shows four frames from the video, reproduced under the MAFW research-use terms. Panel B shows the eleven classifier scores for those frames. Panel C shows smoothed trajectories for the channels with the highest mean scores. Scores were generated with transforEmotion using OpenAI CLIP ViT-Large/14.

the evaluation set. The calibration set determined the LOMM measurement geometry and the dynamic and measurement parameters. LOMM then generated 100 datasets with the same observation positions and uninterrupted segment boundaries as the evaluation videos. We compared LOMM with a static logistic-normal generator calibrated from the same 75 MAFW videos. The static generator retained the estimated compositional structure, between-video heterogeneity, and observation pattern while drawing observations independently over time. This comparison estimates the gain associated with temporal generation beyond static calibration.

For each fold, the MAFW reference comprised 1,000 comparisons between two disjoint, category-stratified groups of 25 videos drawn from the 75 calibration videos. The 22 features were standardized using medians and robust scales estimated from the calibration set. Series similarity was measured by the root-mean-square Euclidean distance across these standardized features. The similarity threshold was the 95th percentile of the directional nearest-neighbor distances in the MAFW reference comparisons. Plausibility is the proportion of simulated series whose nearest MAFW evaluation series fell within this threshold. Coverage is the proportion of MAFW evaluation series whose nearest simulation fell within the threshold. Reported ranges are the 2.5th and 97.5th percentiles across simulation or reference replicates. Appendix B gives the calibration, parameter selection, metric, and reference-sampling procedures.

## Study 2: Recovery of Dimensionality and Indicator Assignments

*Data-generating conditions.* Study 2 asks whether four network-based methods recover a prespecified number of dimensions and a prespecified assignment of indicators to dimensions. The primary analyses use continuous indicators generated by LOMM. Each dataset contains $N$ clips with $T$ observations each and $p = mk$ indicators. All clips in a dataset share the same indicator assignments and the same nominal loadings and residual variances; each clip's loadings and residual variances are perturbed by small random amounts. Datasets are generated under six scenario families (Table 1), which differ in how strongly and how persistently each latent dimension is expressed and in whether the oscillator dynamics are stable or amplifying. Appendix B gives the exact activity schedules and parameters.

Every method receives the same simulated dataset. Each method estimates its own structure, that structure is converted to an undirected network by a prespecified rule, and the same community-detection procedure identifies groups of indicators. The estimated grouping is then compared with the assignment used to generate the data. This creates a common recovery target while preserving each method's own statistical model and data handling.

Table 1: The six Study 2 scenario families. Numeric activity gains, dwell fractions, and oscillator parameters are in Appendix B.

| Code | Family |
|---|---|
| A | Weak expression, one dominant dimension |
| B | Strong expression, one dominant dimension |
| C | Strong expression, switching dominant dimension |
| D | Strong expression, rapid switching |
| E | Amplifying dynamics, one dominant dimension |
| F | Amplifying dynamics, switching dominant dimension |

Table 2: The four Study 2 methods, the structure each supplies for the common grouping step, and its role in the comparison.

| Method | Structure used for grouping | Role in the comparison |
|---|---|---|
| DynEGA | Partial-correlation network of estimated changes | Method of primary interest |
| Static EGA | Partial-correlation network of contemporaneous levels | Ignores temporal order; computationally inexpensive benchmark |
| GraphicalVAR | Contemporaneous residual partial-correlation network | Dynamic network model with lagged and contemporaneous relations |
| GIMME | Average residual-covariance network across clip-level fits | Fitted per clip and averaged; tests computational feasibility |

*Recovery procedures.* DynEGA (Boker et al., 2010; H. Golino et al., 2020; H. Golino et al., 2022; H. F. Golino & Epskamp, 2017) is the method of primary interest. It estimates the first derivative of each indicator from time-delay embeddings, estimates a regularized partial-correlation network among the derivatives, and applies a community-detection algorithm to the network; the communities are the estimated dimensions. GraphicalVAR (Epskamp et al., 2018; Epskamp, 2024) fits a multilevel vector autoregression that separates lagged from contemporaneous relations; its contemporaneous residual partial-correlation network is used for grouping. GIMME (Gates & Molenaar, 2012; Lane et al., 2019) fits a structural vector autoregression to each clip separately; residual covariance matrices from successful fits are averaged across clips, and the average is used for grouping. This use of GIMME is computationally demanding and differs from its intended idiographic application, so the GIMME results apply to this comparison only.

Static EGA estimates a regularized partial-correlation network from the contemporaneous correlation matrix of indicator levels, ignoring the observation order, and applies the same community-detection step. It serves two purposes. Statistically, it shows how much of the grouping is recoverable from contemporaneous levels alone; if most of it is, a dynamic model cannot add much for this recovery target. Computationally, it requires far less time than the dynamic methods. Comparing static EGA with DynEGA and GraphicalVAR therefore shows whether modeling temporal structure improves grouping enough to justify its computational cost in these conditions.

*Primary design and outcomes.* The primary design uses $T = 100$ observations per clip and contains 21,600 datasets. It crosses the six scenario families, $m \in \{2, 3\}$ dimensions, $N \in \{50, 100\}$ clips, three loading levels (0.40, 0.60, 0.80), three residual standard deviations (0.125, 0.25, 0.50), and 100 replications, with $k = 4$ indicators per dimension. Every method receives the same dataset within a condition and replication. Because the MAFW segments have a median length of 35 observations, the same 21,600 conditions were also run with $T = 25$ and are reported as a secondary analysis. Louvain at resolution 1.0 is the primary community-detection algorithm; Leiden and Walktrap were applied to the same fitted networks as sensitivity analyses.

The primary outcome is the overall correct-dimension recovery rate: the proportion of datasets in which the method returned a valid network and the estimated number of dimensions equaled $m$. A failed or timed-out fit counts as an unsuccessful recovery. We also report the valid-output rate and the correct-dimension rate among valid outputs, so that computational feasibility and accuracy can be examined separately. Assignment accuracy, the adjusted Rand index, and variation of information compare the full estimated partition of indicators with the true partition among valid outputs. A secondary diagnostic summarizes each recovered group by its first principal component and compares that summary with the corresponding latent trajectory; these trajectory diagnostics are reported in Appendix C and are not primary outcomes.

*Additional analyses.* Three additional analyses examine the limits of the primary result. First, one-at-a-time sensitivity analyses start from a reference condition ($N = 100$, $T = 100$, loading 0.60, residual SD 0.25, shared initial state) and change one feature: independent rather than shared initial states, gradual rather than abrupt activity transitions, small cross-loadings, residual correlation between indicators, correlated acceleration inputs, or transformation of the indicators to bounded scores. The bounded-score condition applies the softmax function directly to the $p$ continuous indicators and analyzes either the $p$ scores or their centered log-ratio (CLR) components. Because the transformed scores must sum to one, the original block structure among the $p$ variables need not be preserved. Second, a small analysis adds clip-specific baseline and scale differences, with baselines constrained to the column space of the loading matrix so that they add

no structure outside the factor model. Third, an extended factorial simulation of 518,400 datasets evaluates DynEGA alone over additional numbers of dimensions, indicators per dimension, and observation schedules. Appendix B gives the exact settings for every analysis.

# Results

## Study 1 Results: Similarity of Simulated and Real Score Series

Figure 4 compares the two generators with the MAFW reference. LOMM achieved a median plausibility of 0.970 (95% simulation range [0.935, 0.990]) and coverage of 0.920 [0.900, 0.950]. The corresponding MAFW-to-MAFW reference values were 0.950 [0.910, 0.990] for both measures. The static logistic-normal generator achieved plausibility of 0.510 [0.410, 0.590] and coverage of 0.370 [0.290, 0.455]. LOMM therefore generated score series that usually had a close MAFW counterpart and represented most MAFW evaluation series, whereas static calibration alone covered substantially less of the empirical variation.

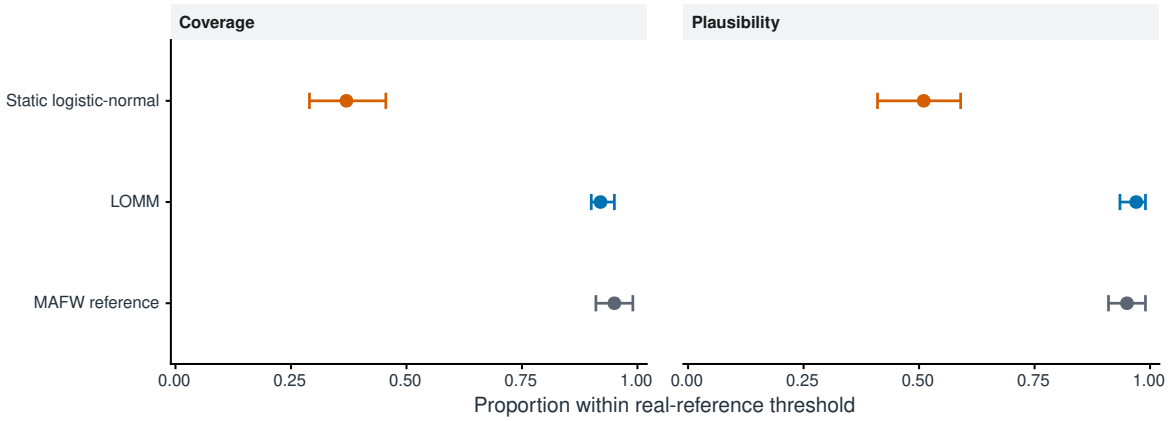



Figure 4: Cross-fitted similarity of LOMM and static logistic-normal series to MAFW evaluation series. Points are median proportions and bars are 2.5th and 97.5th percentiles across 100 generated datasets for each generator or 1,000 MAFW-to-MAFW reference comparisons. Plausibility is the proportion of simulated series with a nearest MAFW evaluation series within the fold-specific threshold. Coverage is the proportion of MAFW evaluation series with a nearest simulation within the same threshold. The threshold is the 95th percentile of directional nearest-neighbor distances between disjoint MAFW calibration samples. Higher values indicate closer agreement.

Across the metric-specific comparisons, 14 of the 22 LOMM feature discrepancies fell within the MAFW-to-MAFW reference range. Agreement was strongest for dispersion, short-term change, temporal persistence, and robust change variance. The clearest remaining differences concerned sample entropy, which was higher in LOMM series, and spectral entropy, which was lower. Appendix C reports all metric-specific results.

## Study 2 Results: Recovery of Dimensionality and Indicator Assignments

Table 3 reports the $T = 100$ results under Louvain. The overall recovery rate requires both a valid network and the correct number of dimensions, so every failure and timeout counts against it. The last two columns report assignment accuracy among valid outputs and median runtime.

Table 3: Results for the primary $T = 100$ design under Louvain. Assignment accuracy is computed among valid network outputs. Runtime is the median, in seconds, among valid network outputs and includes model fitting, network extraction, and community detection.

| Method | Valid output | Correct given valid | Overall recovery | Assignment accuracy | Median runtime, s |
|---|---|---|---|---|---|
| DynEGA | 1.000 | 0.939 | 0.939 | 0.965 | 3.69 |
| Static EGA | 0.997 | 0.887 | 0.884 | 0.922 | 2.15 |
| GraphicalVAR | 0.941 | 0.826 | 0.777 | 0.902 | 85.41 |
| GIMME | 0.224 | 0.783 | 0.176 | 0.877 | 518.16 |

DynEGA returned a valid network for all 21,600 datasets and had the highest overall recovery rate, 0.939. Static EGA followed at 0.884, which shows that most of the grouping was recoverable from contemporaneous indicator levels. GraphicalVAR reached 0.777. GIMME recovered the correct dimension in 0.783 of its valid outputs but returned a valid network for only 0.224 of datasets, so its overall recovery was 0.176. Assignment accuracy among valid outputs was 0.965 for DynEGA, 0.922 for static EGA, 0.902 for GraphicalVAR, and 0.877 for GIMME. Static EGA was about 40 times faster than GraphicalVAR and had higher overall recovery and assignment accuracy for this target. GraphicalVAR estimates a more complex model that includes lagged effects; the comparison evaluates only whether that additional structure improved recovery of the grouping under these conditions. Appendix C reports Monte Carlo standard errors, bootstrap intervals, and the remaining partition diagnostics.

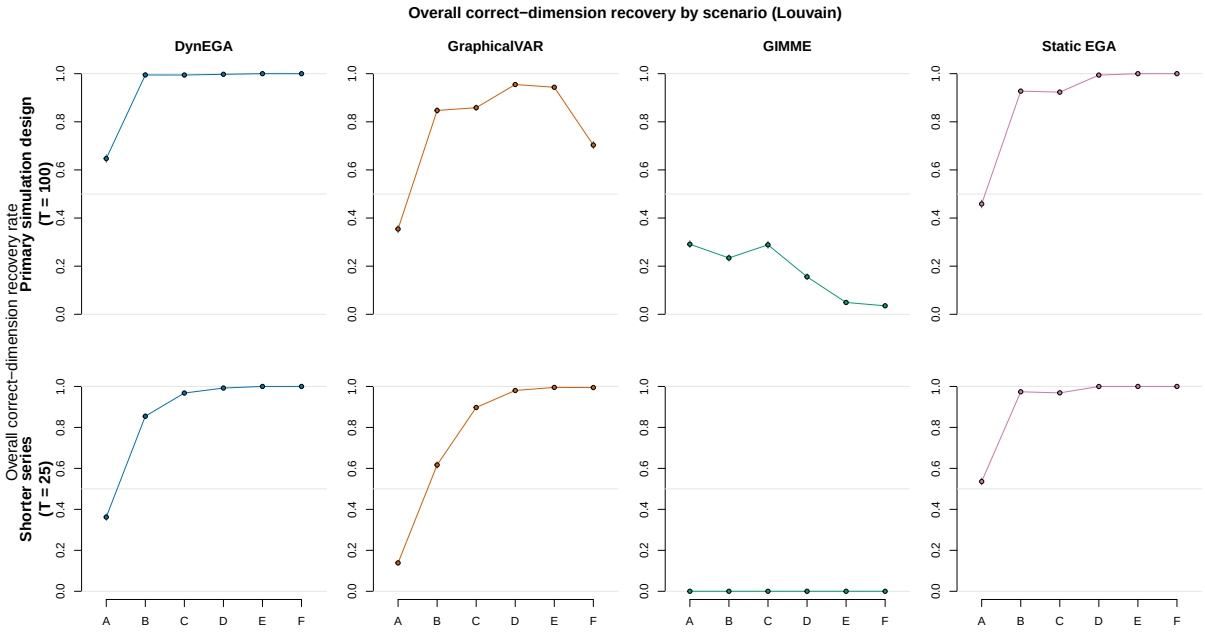



Figure 5: Overall correct-dimension recovery under Louvain by scenario family (codes as in Table 1), method, and observations per clip ($T = 100$, top; $T = 25$, bottom). A recovery requires a valid network output and the correct number of dimensions. Error bars are 95% bootstrap intervals from resampling whole datasets.

Recovery varied across scenario families and observations per clip (Figure 5). DynEGA had the highest recovery in most conditions. Static EGA's recovery was high because the block structure was present in contemporaneous levels. GraphicalVAR's recovery was lowest when expression was weak or measurement error was large. GIMME's low overall rates mainly reflected missing valid outputs. In the extended factorial simulation, DynEGA recovered the correct number of dimensions in 93.7% of 518,400 datasets and assigned 95.2% of indicators correctly; stronger loadings and smaller residual error improved recovery. Leiden and Walktrap led to the same conclusions as Louvain.

As a secondary diagnostic, each recovered group was summarized by its first principal component and compared with the corresponding activity-scaled latent trajectory. Among valid outputs with the correct number of dimensions at $T = 100$, the correlation and normalized RMSE were 0.574 and 1.418 for DynEGA, 0.791 and 0.568 for GraphicalVAR, 0.606 and 0.898 for GIMME, and 0.787 and 0.563 for static EGA. This ordering differs from the ordering for dimensional recovery. Grouping recovery and trajectory recovery are different targets, and neither ordering ranks the methods in general. Appendix C reports the complete diagnostics.

## Robustness and Sensitivity Analyses

*Shorter records.* With $T = 25$, static EGA had the highest overall recovery, 0.913, followed by DynEGA, 0.863, and GraphicalVAR, 0.770. Shortening the record slightly reduced DynEGA's recovery, left GraphicalVAR's about the same, and increased static EGA's. Under the 900-second limit, no GIMME fit at $T = 25$ returned a valid network.

*Initial states.* In the reference condition of the one-at-a-time analyses ($N = 100$, $T = 100$, loading 0.60, residual SD 0.25, shared initial state), overall recovery was 0.993 for DynEGA, 0.912 for static EGA, and 0.683 for GraphicalVAR. Replacing the shared initial state with independent initial states produced the largest change of any sensitivity condition: recovery fell to 0.838, 0.525, and 0.253, respectively. A shared initial state therefore overstates recovery for all three methods.

*Other one-at-a-time changes.* Gradual activity transitions, small cross-loadings, and correlated acceleration inputs changed recovery little. A residual correlation of 0.10 between every pair of indicators increased recovery for GraphicalVAR and static EGA, but because that correlation also links indicators of different dimensions, the increase should not be read as improved performance.

*Bounded scores.* Transforming the $p$ indicators to scores that sum to one changed the variables being analyzed and sharply reduced recovery of the original grouping. DynEGA recovered the correct number of dimensions in 0.333 of datasets analyzed as CLR components and 0.233 analyzed as raw scores; the corresponding rates were 0.333 and 0.167 for static EGA and 0.087 and 0.002 for GraphicalVAR. The softmax transformation and the accompanying projection onto the zero-sum subspace did not preserve the structure among the original indicators. For this reason, the Study 2 recovery results apply to continuous indicators and are not extended to bounded scores of the kind analyzed in Study 1.

*Between-clip baseline and scale.* This analysis used stable families A through D with independent stationary initial states, so its reference recovery rates are lower than in the shared-initial-state analyses. Adding clip-specific baseline and scale changed DynEGA's recovery from 0.775 to 0.750, a paired difference of $-0.025$ (95% interval $[-0.100, 0.050]$). Static EGA rose from 0.275 to 1.000: the added baselines lie in the column space of the loadings, so they add between-clip variation that follows the intended grouping. GraphicalVAR and GIMME were unchanged, but those estimates rest on very few valid outputs; GraphicalVAR produced three correct outputs among 80 datasets, and every three-dimensional GIMME fit timed out.

*Runtime and feasibility.* Median runtime among valid $T = 100$ outputs was 2.15 s for static EGA, 3.69 s for DynEGA, 85.41 s for GraphicalVAR, and 518.16 s for GIMME. GraphicalVAR timed out on 1,271 of 21,600 fits. GIMME produced 4,846 valid outputs, 14,477 timeouts, and 2,277 post-estimation errors. DynEGA

returned a valid output for every fit in the primary and extended designs. These runtimes are specific to our implementation and hardware. Appendix C gives the full runtime, factorial, and community-detection results.

# Discussion

LOMM addresses a common measurement problem in behavioral research. A classifier can produce multivariate time series from video, and an analysis can return dimensions or networks from those series, yet the empirical data alone cannot reveal whether the intended structure was recoverable. LOMM makes the recovery target explicit by specifying latent dynamics, time-varying activity, and the observation model separately. This separation is the paper's central contribution. It also applies beyond video whenever intensive longitudinal measures are treated as fallible indicators of lower-dimensional psychological processes. CLIP is not part of LOMM; it is the scoring pipeline that produced the empirical series in Study 1.

Study 1 found that LOMM generated classifier-score series with high plausibility and broad coverage relative to the MAFW reference. The static logistic-normal generator preserved the calibrated score geometry and between-video heterogeneity. It omitted temporal dependence and performed substantially worse on both measures. Temporal generation was therefore needed to reproduce the joint statistical profile of the MAFW series. The metric-specific results also identify a concrete target for model development: LOMM captured most measures of level, dispersion, change, persistence, and cross-channel variation, while its entropy profile remained less similar to MAFW.

Study 2 evaluated recovery of a known factor structure from continuous indicators. DynEGA had the highest overall recovery and was the only method feasible in the extended factorial simulation. Static EGA showed that most of the grouping was recoverable from contemporaneous levels; it exceeded GraphicalVAR in both dimensional recovery and assignment accuracy while running in a median of 2.15 s rather than 85.41 s. The practical implication is that, for this recovery target and these conditions, modeling temporal structure did not improve recovery enough to offset its computational cost. GraphicalVAR answers additional questions about lagged relations, which this comparison does not evaluate.

The trajectory diagnostics in Appendix C show that the ranking of methods depends on the recovery target. DynEGA's advantage concerned recovery of the number of dimensions and the indicator partition; alignment of recovered-group summaries with the latent trajectories produced a different ordering. The two targets should be evaluated separately. All comparisons apply to the specific estimation, network-conversion, and grouping procedures used here. GIMME's low overall recovery mainly reflected its failure to return valid

outputs under a clip-level fitting procedure that differs from its usual use.

The most important qualification of the primary result concerns initial states. A shared starting state creates a shared initial transient across dimensions, which adds recoverable structure that is not part of the measurement partition. Independent initial states reduced recovery for DynEGA, static EGA, and GraphicalVAR, so the shared-initial-state design overstated recovery. Gradual activity transitions, small cross-loadings, and correlated acceleration inputs changed little. The bounded-score condition set a clearer limit: the softmax transformation changed the associations among the original indicators, and neither the raw scores nor the CLR components preserved the original grouping. The Study 2 recovery claims therefore apply to continuous indicators.

*Limitations.* Several limitations remain. The latent dimensions are uncoupled. The amplifying families produce large amplitudes late in the record and test recovery under those conditions only. The $T = 300$ conditions also change the observation interval, so they do not isolate the number of observations per clip. The GIMME residual-covariance conversion was constructed for this comparison and differs from typical idiographic use. Study 1 uses 100 short MAFW videos, one zero-shot scoring pipeline, and one LOMM specification chosen during model development on the same sample. LOMM is used here only as a generator with prespecified parameters; estimating it from data would require the identification constraints listed in Appendix A.

Three practical conclusions follow. Initial conditions can create structure unrelated to the intended measurement partition. Transforming indicators to bounded scores can change the structure among the analyzed variables. Computational feasibility belongs in any recovery evaluation, because a method that returns no output cannot recover the target. Applied researchers should therefore specify the latent dynamics, activity process, observation model, and recovery target before interpreting an empirical network. LOMM supports this workflow and can be extended to other scoring systems, coupled latent dimensions, and recovery targets defined directly for bounded data.

# Declarations

## Funding

The authors received no specific funding for this work.

## Competing interests

The authors declare that they have no competing interests.

## Ethics approval

This study used simulations and secondary video data obtained under the MAFW research-use terms. It involved no new recruitment, intervention, or direct collection of human-participant data by the authors.

## Consent to participate

Not applicable.

## Consent for publication

The manuscript includes four still frames from MAFW solely to illustrate the score-extraction procedure. The frames are cited to the dataset, and no source video is redistributed. Their inclusion follows the dataset's terms for limited reproduction in academic publications.

## Availability of data and materials

The project GitHub repository at https://github.com/atomashevic/LOMM contains the simulation summary results, analysis-ready tables underlying the figures, and aggregate MAFW-derived summaries used in the simulated-versus-real comparison. Raw MAFW videos and frame-level classifier scores are not redistributed. Access to MAFW is governed by the dataset provider, and an end-to-end reanalysis of Study 1 requires approved access and local generation of the classifier scores.

## Code availability

Code for generating LOMM datasets, fitting the methods, reproducing the statistical summaries, and regenerating the tables and figures is available in the same GitHub repository. The repository also provides the manuscript source and instructions for reproducing the analyses.

## Authors' contributions

Aleksandar Tomašević contributed conceptualization, formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Hudson Golino contributed formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Alexander P. Christensen contributed methodology, software, writing the original draft, and reviewing and editing the manuscript.

# Open Practices Statement

The study was not preregistered. The simulation and analysis code, manuscript source, paper-facing tables, and aggregate MAFW-derived summaries used in the simulated-versus-real comparison are available at https://github.com/atomashevic/LOMM. Raw MAFW videos are third-party materials, and the frame-level classifier scores are derived from those restricted materials; neither is included.

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# Appendix A: Complete Specification of LOMM

This appendix gives the formal LOMM specification behind the conceptual account in the main text. It is intended for readers who want to reimplement the generator or examine its assumptions and identification limits.

## A.1 The Latent Transition

Equation 1 gives the continuous-time oscillator. The implementation advances its two-component state exactly over one observation interval using

$$\Phi_f(\Delta) = \exp(\mathbf{A}_f\Delta), \qquad \mathbf{A}_f = \begin{bmatrix} 0 & 1 \\ \eta_f & \zeta_f \end{bmatrix} \qquad \text{and} \qquad \Gamma_f(\Delta) = \int_0^\Delta \exp(\mathbf{A}_f u)\mathbf{b}\, du, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

For the nonzero position-feedback values used in the simulation grids, the input map has the equivalent exact form

$$\Gamma_f(\Delta) = \mathbf{A}_f^{-1}\{\Phi_f(\Delta) - \mathbf{I}_2\}\mathbf{b}.$$

The integral definition remains valid when $\mathbf{A}_f$ is singular, including $\eta_f = 0$. If a root-based closed form is used numerically, a separate repeated-root branch is required when $\zeta_f^2 + 4\eta_f$ is close to zero.

The state components $x_{if,n}$ and $v_{if,n}$ are the level and its rate of change at $t_n$; $\eta_f$ is position feedback, $\zeta_f$ is velocity feedback, and $\sigma_{qf}$ is the standard deviation of the acceleration input. On the interval $[t_n, t_{n+1})$ the underlying continuous-time model is $\ddot{x}_{if}(t) = \eta_f x_{if}(t) + \zeta_f \dot{x}_{if}(t) + q_{if,n}$ with $q_{if,n}$ held constant, so $\Phi_f(\Delta)$ propagates the deterministic state and $\Gamma_f(\Delta)$ loads the input. The inputs are independent across videos, dimensions, and transitions, and independent of initial states, activity schedules, and measurement errors. They are zero-order-hold acceleration inputs rather than increments of a continuous-time white-noise diffusion. The covariance added in one transition is $\Gamma_f(\Delta)\sigma_{qf}^2\Gamma_f(\Delta)^\top$.

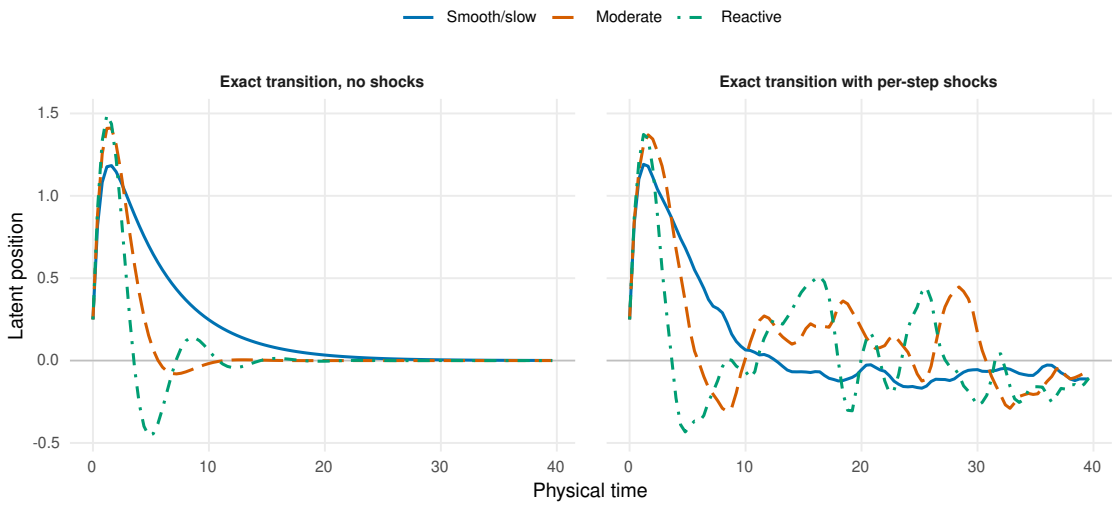



Figure A1: Illustrative latent trajectories for three uncoupled dimensions under the exact transition, shown without and with piecewise-constant acceleration inputs. The inputs create irregular peaks and reversals while the feedback parameters govern the underlying dynamic regime.

## A.2 Dynamic Regimes and Finite-Horizon Amplification

The characteristic roots for dimension $f$ are

$$r_{f,\pm} = \frac{\zeta_f \pm \sqrt{\zeta_f^2 + 4\eta_f}}{2}.$$

Continuous-time asymptotic stability requires $\eta_f < 0$ and $\zeta_f < 0$, and the roots are oscillatory when $\zeta_f^2 + 4\eta_f < 0$. The exact discrete transition is asymptotically stable when

$$\rho\{\Phi_f(\Delta)\} = \exp\left[\Delta \max\{\mathrm{Re}(r_{f,+}), \mathrm{Re}(r_{f,-})\}\right] < 1.$$

Negative $\eta_f$ supplies restoring acceleration and negative $\zeta_f$ attenuates velocity, whereas positive $\zeta_f$ produces amplification. How much amplification actually appears depends on the observed horizon. For $T$ observations, $H = (T-1)\Delta$. Let $\alpha(\mathbf{A}_f) = \max\{\mathrm{Re}(r_{f,+}), \mathrm{Re}(r_{f,-})\}$. The quantity

$$g_f(H) = \exp\{H\alpha(\mathbf{A}_f)\}$$

is a modal exponential-rate factor over that horizon. It does not by itself measure nonnormal transient amplification or the polynomial factor that can occur at a defective repeated root. A norm-based finite-

horizon state-amplification diagnostic is

$$G_f^{\max}(H) = \sup_{0 \le u \le H} \| \exp(\mathbf{A}_f u) \|_2,$$

and the corresponding level-only diagnostic is

$$G_{x,f}^{\max}(H) = \sup_{0 \le u \le H} \left\| \begin{bmatrix} 1 & 0 \end{bmatrix} \exp(\mathbf{A}_f u) \right\|_2 .$$

The Euclidean matrix norm is coordinate-dependent. Because the state combines a level and a rate of change, whose components have different units, this diagnostic is conditional on the chosen state coordinates and should be interpreted only after fixing a nondimensionalization or an explicit weighted state norm. A coordinate-aware definition is

$$G_{f,M}^{\max}(H) = \sup_{0 \le u \le H} \left\| \mathbf{M}^{1/2} \exp(\mathbf{A}_f u) \mathbf{M}^{-1/2} \right\|_2 ,$$

for a stated positive-definite state metric $\mathbf{M}$; the unweighted $G_f^{\max}(H)$ corresponds to $\mathbf{M} = \mathbf{I}_2$. Reporting the roots and $g_f(H)$ characterizes the modal regime over the observed horizon, but $g_f(H)$ should not be read as total finite-horizon amplification. Reporting the activity schedule separately keeps amplification in the latent dynamics distinct from nonstationarity created by the activity layer.

## A.3 Initial States

The system is uncoupled: $\eta_f$, $\zeta_f$, and $\sigma_{qf}$ may vary by dimension, and off-diagonal dynamic effects are omitted. Study 2's primary simulation design initialized every dimension at $x_{if,1} = 0.25$ and $v_{if,1} = 2$. The sensitivity analysis used independent starting states. For a stable dimension, the stationary covariance $\mathbf{P}_f$ solves

$$\mathbf{P}_f = \Phi_f(\Delta) \mathbf{P}_f \Phi_f(\Delta)^\top + \Gamma_f(\Delta) \sigma_{qf}^2 \Gamma_f(\Delta)^\top,$$

whose solution is unique when $\rho\{\Phi_f(\Delta)\} < 1$. Drawing $\mathbf{s}_{if,1} \sim \mathcal{N}(\mathbf{0}, \mathbf{P}_f)$ makes the state process stationary before the activity layer is applied; time-varying activity modulation still renders $a_{if}(t) x_{if}(t)$ nonstationary. For a dimension with positive velocity feedback, the sensitivity draws $x_{if,1} \sim \text{Unif}(-0.25, 0.25)$ and $v_{if,1} \sim \text{Unif}(-2, 2)$. These independent starts remove the transient that all dimensions would otherwise share, while

leaving the observed horizon unchanged.

## A.4 The Activity Layer at a Transition

Away from a discontinuity in the activity value, the ordinary product rule applies,

$$\frac{d}{dt}\{a_{if}(t)x_{if}(t)\} = a'_{if}(t)x_{if}(t) + a_{if}(t)x'_{if}(t).$$

At an abrupt transition, the derivative of the activity value contains an impulsive component in the distributional sense. Define $D_n x = x_{n+1} - x_n$. The exact discrete identity is

$$D_n(a_{if}x_{if}) = a_{if,n+1}D_n x_{if} + (D_n a_{if})x_{if,n},$$

where the first term uses the post-transition activity value and the second uses the pre-transition state. The corresponding finite-difference identity is

$$\frac{\tilde{x}_{if,n+1} - \tilde{x}_{if,n}}{\Delta} = a_{if,n+1}\frac{x_{if,n+1} - x_{if,n}}{\Delta} + \frac{a_{if,n+1} - a_{if,n}}{\Delta}x_{if,n}.$$

The second term is the term that enters an estimated derivative at an activity switch even when the oscillator itself has not accelerated; this motivates the gradual-transition sensitivity analysis. This convention attaches the new activity value to the post-transition observation, so the activity layer multiplies each completed latent state by the schedule value at the same observation index.

The activity schedule takes two nonnegative values, $a_{\text{high}}$ and $a_{\text{low}}$. In non-switching families the dominant dimension remains at $a_{\text{high}}$ for the whole record and the others remain at $a_{\text{low}}$. In switching families, dwell fractions determine how long each dimension remains dominant before the schedule reassigns $a_{\text{high}}$. Appendix B lists the numeric gains and dwell values used in the Study 2 families.

## A.5 The Observation Layer

Let $J$ denote the number of positive score channels, let $d = J - 1$, and let $\mathbf{V} \in \mathbb{R}^{J\times d}$ be a fixed orthonormal isometric log-ratio (ILR) basis satisfying $\mathbf{V}^\top\mathbf{V} = \mathbf{I}_d$ and $\mathbf{V}^\top\mathbf{1}_J = \mathbf{0}$. The Study 1 basis used here is a normalized Helmert basis constructed in the stated 11-channel order. The full observation model in log-ratio coordinates is

$$\mathbf{u}_{in} = \mu + \mathbf{b}_i + \kappa s_i \left(\Lambda_i \tilde{\mathbf{x}}_{in} + \varepsilon_{in}\right).$$

Here $\mu \in \mathbb{R}^d$ is a fixed center, $\mathbf{b}_i \in \mathbb{R}^d$ is the clip baseline, $s_i > 0$ is the clip scale, $\kappa > 0$ is a global scale, and $\Lambda_i \in \mathbb{R}^{d \times m}$ maps the $m$ activity-scaled latent positions into these coordinates. The clip scale multiplies both signal and residual, whereas the baseline is additive. Conditional on the clip-specific terms and $\Sigma_i$, measurement errors are independent over time and across clips, and independent of latent states, initial states, activity schedules, and acceleration inputs, with $\varepsilon_{in} \mid \Sigma_i \sim \mathcal{N}(\mathbf{0}, \Sigma_i)$. Dependence across coordinates follows $\Sigma_i$. Designs with fixed measurement parameters set $\Lambda_i = \Lambda$ and $\Sigma_i = \Sigma$ for every clip. The residual-correlation sensitivity condition sets $\Sigma = s_\varepsilon^2\{(1-\rho)\mathbf{I}_p + \rho \mathbf{1}_p \mathbf{1}_p^\top\}$ with $\rho = 0.10$ across all $p$ indicators, thereby introducing residual covariance both within and between the prespecified blocks.

The three baseline constructions differ only in the subspace from which $\mathbf{b}_i$ is drawn. Study 1 draws it from a zero-mean Gaussian whose covariance is estimated from MAFW calibration videos in the full coordinate space. Study 2's between-clip-variability analysis instead uses

$$\mathbf{b}_i = \Lambda \beta_i, \qquad E(\beta_i) = \mathbf{0},$$

so that between-clip variation stays within the space spanned by the loadings and does not introduce structure outside the factor model. Study 2 conditions without baseline and scale differences fix $\mathbf{b}_i = \mathbf{0}$ and $s_i = 1$, giving

$$\mathbf{u}_{in} = \mu + \kappa \left(\Lambda_i \tilde{\mathbf{x}}_{in} + \varepsilon_{in}\right).$$

In the matched-method and extended factorial designs, $\Lambda_i$ and the diagonal of $\Sigma_i$ receive small clip-specific perturbations to the loadings and residual variances. The one-at-a-time sensitivity and between-clip-variability designs use fixed equal values except where the stated condition changes them. Thus $\mathbf{b}_i = \mathbf{0}$ and $s_i = 1$ remove baseline and overall-scale differences; they do not by themselves imply a common loading matrix or residual covariance.

The continuous indicators used in Study 2's primary recovery designs are the indicator vector before any bounded-score mapping,

$$\mathbf{z}_{in} = \Lambda_i \tilde{\mathbf{x}}_{in} + \varepsilon_{in},$$

in which each of the $p = mk$ indicators loads on one dimension and has diagonal measurement error.

For bounded output, LOMM maps the coordinates to the open probability simplex. The centered log-ratio (CLR) expresses each channel relative to the geometric mean of all channels. The isometric log-ratio (ILR) represents the same information in $d = J - 1$ orthonormal coordinates. Formally,

$$\mathrm{clr}(\mathbf{p}) = \log \mathbf{p} - J^{-1}\{\mathbf{1}_J^\top \log(\mathbf{p})\}\mathbf{1}_J, \qquad \mathrm{ilr}_{\mathbf{V}}(\mathbf{p}) = \mathbf{V}^\top \mathrm{clr}(\mathbf{p}),$$

with the logarithm applied componentwise. For the canonical bounded LOMM output, applying softmax to $\mathbf{V}\mathbf{u}_{in}/\tau$ gives

$$\mathrm{ilr}_{\mathbf{V}}(\mathbf{p}_{in}) = \frac{\mathbf{u}_{in}}{\tau}, \qquad \mathrm{clr}(\mathbf{p}_{in}) = \frac{\mathbf{V}\mathbf{u}_{in}}{\tau},$$

the second because $\mathbf{V}\mathbf{u}_{in}$ lies in the zero-sum log-ratio subspace. The fixed basis excludes the unidentified common-logit direction and the softmax produces strictly positive rows summing to one, so the scores identify the scaled coordinates $\mathbf{u}_{in}/\tau$ exactly. A common-logit component remains unobservable, and the unscaled coordinates and $\tau$ are not separately identified without an external scale constraint.

The Study 2 bounded-score sensitivity condition uses the $p = mk$ continuous-indicator vector directly. With $J = p$, it computes

$$\mathbf{p}_{in}^{\mathrm{dep}} = \mathrm{softmax}\left(\frac{\mathbf{z}_{in}}{\tau}\right), \qquad \mathrm{clr}\big(\mathbf{p}_{in}^{\mathrm{dep}}\big) = \frac{1}{\tau}\mathbf{P}_p\mathbf{z}_{in}, \qquad \mathbf{P}_p = \mathbf{I}_p - \frac{\mathbf{1}_p\mathbf{1}_p^\top}{p}.$$

For any orthonormal ILR basis $\mathbf{V}_p \in \mathbb{R}^{p\times(p-1)}$, $\mathbf{V}_p\mathbf{V}_p^\top = \mathbf{P}_p$. Defining $\mathbf{u}_{in}^{\mathrm{dep}} = \mathbf{V}_p^\top\mathbf{z}_{in}$ therefore gives $\mathbf{V}_p\mathbf{u}_{in}^{\mathrm{dep}} = \mathbf{P}_p\mathbf{z}_{in}$. Softmax is invariant to the removed row mean, so the direct-softmax branch is equivalent to applying the basis form to $\mathbf{u}_{in}^{\mathrm{dep}}$. The analysis used the $p$ raw scores and the $p$ CLR components, which satisfy a zero-sum constraint; it did not analyze the $(p-1)$ ILR coordinates. The common-logit direction in $\mathbf{z}_{in}$ is unidentifiable after this transformation.

## A.6 Converting Period and Damping to Feedback Parameters

Study 1's search was parameterized by natural period and damping ratio rather than by $\eta_f$ and $\zeta_f$ directly. If $P_f$ is the natural period in observation intervals and $\xi_f$ is the damping ratio, then

$$\omega_{n,f} = \frac{2\pi}{P_f \Delta}, \qquad \eta_f = -\omega_{n,f}^2, \qquad \zeta_f = -2\xi_f \omega_{n,f},$$

and the damped angular frequency is $\omega_{d,f} = \omega_{n,f}\sqrt{1-\xi_f^2}$. With the Study 1 settings $P_f = 12$, $\xi_f = 0.20$, and $\Delta = 1$, these give $\eta_f = -0.2742$ and $\zeta_f = -0.2094$. The acceleration inputs were scaled to give unit stationary position variance.

## A.7 Identifiability

For any nonsingular diagonal matrix $\mathbf{D}$,

$$\Lambda \tilde{\mathbf{x}}_{in} = (\Lambda \mathbf{D}^{-1})(\mathbf{D}\tilde{\mathbf{x}}_{in}),$$

so loading magnitude and latent-trajectory scale are not jointly identified without an explicit scale constraint. Activity gain and acceleration-input scale contribute to latent amplitude in the same way. The softmax layer adds invariance to a common location shift and a tradeoff between the temperature and the coordinate scale. Dimension labels are recoverable only up to permutation.

Sampling cadence adds an aliasing restriction. For an underdamped dimension, write

$$r_{f,\pm} = \frac{\zeta_f}{2} \pm i\omega_f, \qquad \omega_f = \frac{1}{2}\sqrt{-(\zeta_f^2 + 4\eta_f)}.$$

At observation times $t_n = (n-1)\Delta$, the frequencies $\omega_f$ and $\omega_f + 2\pi q/\Delta$ for integer $q$ generate the same sampled sine and cosine sequences after an appropriate change in initial velocity. Identifying continuous-time oscillator parameters from data therefore requires a cadence-dependent restriction such as

$$0 < \omega_f < \frac{\pi}{\Delta},$$

together with a specified acceleration-input model and constraints on the latent state basis. Estimating LOMM from data would in total require sign and scale constraints on the dimensions, normalization of the activity schedule, anchoring of either the loading scale or the input scale, a reference-logit or sum-to-zero constraint, a fixed or otherwise identified temperature, and this frequency restriction. Every oscillatory mode used here satisfies the restriction: Study 1 uses $\Delta = 1$, giving the most restrictive bound $\pi \approx 3.14$ against a largest angular frequency of approximately 0.78, and Study 2 uses $\Delta \in \{0.25, 0.4\}$, giving bounds

of approximately 12.57 and 7.85 against a largest angular frequency of approximately 1.04. The simulations do not estimate any of these parameters.

# Appendix B: Study Designs and Implementation Details

This appendix gives the design and implementation details for both studies. Study 1 evaluates resemblance to real bounded-score series. Study 2 evaluates recovery of a prespecified factor structure from continuous indicators.

## B.1 Study 1 Scoring and Calibration

The sample contains 100 MAFW videos selected across the available expression categories. Category labels served only to stratify sampling and fold construction. The analysis retained 3,557 complete observations in 188 uninterrupted segments. Individual videos contain 16 to 50 complete observations, with a median of 35. Segment lengths range from 1 to 50 observations, with a median of 13.5.

transforEmotion sampled up to 50 frames uniformly from each video and used OpenAI CLIP ViT-Large/14. A Haar cascade detected faces, and the largest detected face was expanded by 50 pixels on each side before classification. Frames without a detected face received missing scores. Eleven lowercase labels were passed directly to the text encoder: anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, helplessness, and disappointment. Burned-in subtitles were not masked when they fell inside the padded face crop.

Scores were required to lie in $[0, 1]$. A complete row was retained when its sum was positive, finite, and within $10^{-6}$ of one, then divided by that sum in the fixed channel order. Values were not truncated, and zero replacement was disabled. The analysis used complete observations and preserved the uninterrupted segment boundaries.

The evaluation panel contains eleven statistics. Median level, interquartile range of levels, interquartile range of observation-to-observation changes, lag-1 autocorrelation, and run-length, sample, permutation, and spectral entropy were computed within each channel. The median across channels summarized each statistic for a video. Sample entropy used embedding order 2 and a tolerance of 0.20 within-series standard deviations. Permutation entropy used order 3. Effective rank was computed from the cross-channel covariance eigenvalues. Robust level variance was the 10% trimmed mean squared distance from the video center divided by the coordinate dimension. Robust change variance was the 10% trimmed mean squared within-segment first difference divided by twice the coordinate dimension. Each statistic was computed on the eleven score channels and on their ten isometric log-ratio coordinates, producing 22 features per series.

The 100 videos were divided into four category-stratified folds. Each fold used 75 calibration videos and 25 evaluation videos. The calibration videos determined the coordinate center, loading subspace, between-

video baseline covariance, and empirical distribution of within-video scale. The first three eigenvectors of the pooled within-video covariance supplied the loading directions. The between-video baseline covariance was shrunk halfway toward its diagonal, and its eigenvalues were floored at $10^{-8}$. The generator included between-video baseline and scale variation and used abrupt activity schedules.

Generator settings were selected within each calibration set. The dynamic search crossed periods of 12, 24, and 48 observations with damping ratios of 0.01, 0.05, and 0.20. Selection used the interquartile range of changes, lag-1 autocorrelation, run-length entropy, and spectral entropy. Conditional on the selected dynamics, the measurement search crossed residual standard deviations of 0.10, 0.20, and 0.30 with $\kappa$ values of 0.75, 1.00, and 1.25. Selection used the full discrepancy panel. Each candidate was evaluated with three calibration simulations. All four folds selected a period of 12 observations, damping ratio 0.20, residual standard deviation 0.30, and $\kappa = 0.75$.

The static logistic-normal generator used the same calibrated coordinate center, covariance geometry, between-video baseline and scale distributions, observation positions, and segment boundaries. It drew independent log-ratio observations within each video. Each selected generator produced 100 datasets per fold with 25 simulated series in each dataset.

The MAFW reference used 1,000 comparisons per fold. Each comparison drew two disjoint, category-stratified groups of 25 videos from the 75 calibration videos. Every feature was standardized by the median and robust scale of the calibration videos. The distance between two series was the root-mean-square Euclidean distance across the 22 standardized features. The fold-specific similarity threshold was the 95th percentile of the pooled directional nearest-neighbor distances between the two MAFW groups.

For each generated dataset, plausibility was the proportion of its 25 simulated series whose nearest MAFW evaluation series fell within the threshold. Coverage was the proportion of the 25 MAFW evaluation series whose nearest simulated series fell within the threshold. Results were averaged across the four folds. The reported ranges are the 2.5th and 97.5th percentiles across the 100 generated datasets or 1,000 MAFW reference comparisons.

Metric-specific distributional discrepancy was computed separately for each of the 22 features. The absolute median difference and one-dimensional Wasserstein distance were divided by the robust calibration scale, and the larger standardized value was retained. The full distributional discrepancy is the mean across the 22 features.

Table B1: The six Study 2 scenario families. Dwell values are fractions of the number of observations per clip.

| Code | Family | Switching | Active / background gain | Dwell fraction | Dynamic rule |
|---|---|---|---|---|---|
| A | Weak expression, one dominant dimension | No | 0.35 / 0.05 | 1.00 / 1.00 | Stable, low input scale |
| B | Strong expression, one dominant dimension | No | 1.00 / 0.08 | 1.00 / 1.00 | Stable, standard input scale |
| C | Strong expression, switching dominant dimension | Yes | 1.00 / 0.08 | 0.20 / 0.45 | Stable, standard input scale |
| D | Strong expression, rapid switching | Yes | 1.25 / 0.08 | 0.03 / 0.10 | Stable, high input scale |
| E | Amplifying dynamics, one dominant dimension | No | 1.00 / 0.08 | 1.00 / 1.00 | Positive velocity feedback |
| F | Amplifying dynamics, switching dominant dimension | Yes | 1.10 / 0.10 | 0.12 / 0.30 | Positive velocity feedback |

## B.2 Study 2 Simulation Designs

For each clip, LOMM sets or draws the starting state and acceleration inputs according to the configuration, advances each latent dimension, applies the activity schedule, and constructs the observed indicators. The primary, shorter-series, and extended factorial designs use continuous indicators. All clips share the same indicator assignments and the same nominal loadings and residual variances; each clip's loadings and residual variances are perturbed by small random amounts. The one-at-a-time sensitivity conditions use fixed equal measurement parameters except for the stated change.

In non-switching scenarios, one dimension remains dominant for a whole clip and the dominant dimension rotates across clips. Switching scenarios sample dwell intervals and change the dominant dimension over time. Families A through D use stable dynamics. Families E and F use positive velocity feedback and test recovery under finite-horizon amplification.

The primary $T = 100$ design contains 21,600 datasets. It crosses six scenario families, $m \in \{2, 3\}$ dimensions, $N \in \{50, 100\}$ clips, loadings $\{0.40, 0.60, 0.80\}$, residual standard deviations $\{0.125, 0.25, 0.50\}$, four indicators per dimension, and 100 paired replications. The shorter-series design repeats the same 21,600 datasets with $T = 25$, $\Delta = 0.4$, and duration 9.6. Every method receives the same dataset within each condition and replication.

The one-at-a-time sensitivity analyses fix $N = 100$, $T = 100$, loading 0.60, residual standard deviation 0.25, and four indicators per dimension (the reference condition). They cross all six families, two dimensionalities, and 50 paired replications while changing one feature at a time.

The between-clip analysis uses 80 paired datasets from stable families A through D, $m \in \{2, 3\}$, and 10 replications. Each dataset has $N = 100$, $T = 100$, four indicators per dimension, loading 0.60, residual standard deviation 0.25, and independent stationary starts. Its baseline varies within the loading space, and its scale varies across clips.

Table B2: One-at-a-time sensitivity conditions. Each changes one feature of the reference condition.

| Sensitivity condition | Specification |
|---|---|
| Independent starting states | Stationary zero-mean starts for stable dimensions; bounded independent starts for amplifying dimensions |
| Gradual activity transitions | A five-observation transition replaces each abrupt switch |
| Cross-loadings | Small normal cross-loadings, truncated to [-0.10, 0.10] |
| Residual equicorrelation | Correlation 0.10 between every residual pair, including pairs from different dimensions |
| Correlated inputs | Acceleration-input correlation 0.15 between dimensions |
| Bounded scores | Direct softmax of the p indicators with tau = 1, analyzed as raw scores or CLR components |

The extended factorial simulation evaluates DynEGA on 518,400 datasets. It crosses $N \in \{50, 100\}$, $T \in \{100, 300\}$, $m \in \{2, 3, 4\}$, $k \in \{4, 6\}$, three loading levels, three residual-error levels, six scenarios, and 400 replications. The $T = 100$ conditions use $\Delta = 0.4$ and the $T = 300$ conditions use $\Delta = 0.25$ while holding per-step input variance fixed. Their difference therefore combines the number of observations, the observation interval, the physical duration, and the discrete input specification.

The public lomm_observe() function implements the observation step and records the configuration, seeds, and hashes needed for reproduction. Study 2 uses the benchmark generator documented in the analysis code, which returns continuous indicators for the primary designs and applies the direct-softmax and CLR transformation for the bounded-score sensitivity condition.

## B.3 Method Implementations

DynEGA estimates first derivatives from time-delay embeddings, constructs an EBICglasso partial-correlation network, and groups its positive edges (Boker et al., 2010; H. Golino et al., 2020; H. Golino et al., 2022; H. F. Golino & Epskamp, 2017). Video identity prevents embedding windows from crossing clip boundaries. An explicit search selects the embedding dimension by the Total Entropy Fit Index, with shorter embeddings breaking ties. Candidate dimensions are $\{5, 10, 15, 25, 50\}$ for $T = 100$ and $\{5, 6, 10, 12\}$ for $T = 25$. EGAnet 2.3.0 uses GLLA embedding parameters tau = 1 and delta = 1, first derivatives, automatic correlations, pairwise missing-data handling, and the glasso model. The derivative scale does not affect the correlation matrix because conversion from observation units to physical time multiplies every derivative by the same positive constant.

Static EGA applies EBICglasso to the contemporaneous correlation matrix of the stacked indicator levels and then uses the same network-processing and community-detection rules as DynEGA. It ignores the observation order. This makes it a benchmark that shows how much of the grouping is recoverable without temporal information, and a computational benchmark for the temporal procedures.

GraphicalVAR uses mlGraphicalVAR with clip identity as the grouping variable, within-clip centering, vari-

able standardization, and gamma 0.5 (Epskamp et al., 2018; Epskamp, 2024). Lagged pairs never cross clip boundaries. The grouping uses the returned fixedPCC contemporaneous residual partial-correlation matrix. The diagonal is removed and the matrix is treated as an undirected weighted graph.

GIMME fits each standardized clip separately with unified structural equation modeling and vector autoregression (Gates & Molenaar, 2012; Lane et al., 2019). Standardized residual covariance matrices from successful fits are aligned by variable name, averaged across clips, and symmetrized. The resulting graph is a residual-covariance graph. Unequal residual variances can therefore influence its weights. This conversion was built for the present comparison and differs from typical idiographic use.

For every method, negative graph weights are set to zero. If no positive edge remains, absolute values of the original off-diagonal weights are used; only an all-zero matrix produces singleton communities. The integrated outputs do not record the frequency of this fallback. Louvain at resolution 1.0 is primary (Blondel et al., 2008). Leiden and Walktrap are applied to the same fitted network as sensitivity analyses (Pons & Latapy, 2005; Traag et al., 2019). A deterministic offset from the dataset seed initializes each method fit.

GraphicalVAR and GIMME were omitted from the 518,400-dataset extended factorial simulation because the full design was computationally impractical. Both were included in the primary and shorter-series designs. GraphicalVAR was also included in the one-at-a-time sensitivity and between-clip analyses. GIMME was included in the between-clip analysis and excluded from the one-at-a-time sensitivity analyses.

## B.4 Outcomes, Uncertainty, and Computing

The valid-output rate records whether a procedure returned a network that could be grouped. The conditional correct-dimension rate records correct dimensionality among valid outputs. The primary outcome, the overall correct-dimension recovery rate, requires both events, so errors and timeouts remain in its denominator.

Adjusted Rand index and variation of information compare the full estimated and prespecified partitions among valid outputs. Assignment accuracy matches estimated groups to dimensions to maximize correct indicator assignments. Every indicator remains in the denominator, and indicators in unmatched groups count as incorrect.

Trajectory-summary diagnostics are constructed after grouping. Indicators are standardized and each estimated group is summarized by its first principal component, with a row mean used when a component cannot be computed. Correlation pairs each prespecified dimension with the estimated summary having the largest absolute correlation. NRMSE uses one-to-one matching after sign reversal when needed. An

unmatched data-generating dimension receives the penalty 2. All-attempt failure-adjusted versions assign correlation 0 and NRMSE 2 when no valid network output exists.

Uncertainty uses 2,000 bootstrap resamples of whole datasets, stratified by condition. Monte Carlo standard errors use delete-one-dataset jackknifing. The three community-detection algorithms reuse the same fitted structure and therefore assess the grouping step rather than estimator variability.

Analyses used R 4.5.1 on Rocky Linux 8.8 with dual Intel Xeon Silver 4310 processors, 48 logical threads, and 125 GiB RAM. Key versions were EGAnet 2.3.0, graphicalVAR 0.3.4, gimme 10.0, igraph 2.2.1, qgraph 1.9.8, psych 2.5.6, lavaan 0.6-20, dplyr 1.1.4, and ggplot2 4.0.1. GraphicalVAR and GIMME used a 900-second limit per fit in the primary and shorter designs.

Runtime is elapsed wall-clock time for attempts returning a valid network output. It includes structure estimation, network extraction, and community detection. One fitted structure is reused for Louvain, Leiden, and Walktrap, so feasibility counts unique fits and accuracy counts fit-and-algorithm results. Runtime values are specific to this implementation and machine.

# Appendix C: Supplementary Results

## C.1 Metric-Level Results for Study 1

Figure C1 compares LOMM with the MAFW-to-MAFW reference for each of the 22 features. Fourteen LOMM discrepancies fell within the reference range. The full 22-feature distributional discrepancy was 0.2543 [0.2356, 0.2738] for LOMM and 0.1811 [0.1521, 0.2193] for the MAFW reference. The corresponding median for the static logistic-normal generator was 0.6942.

LOMM closely reproduced most measures of level, dispersion, short-term change, persistence, effective dimensionality, and robust variation. Its largest departures occurred for entropy. Sample entropy was higher and spectral entropy was lower than in MAFW for both scores and log-ratio coordinates. Log-ratio permutation entropy was lower, effective rank was higher, and robust level variance was lower. The score median-level discrepancy was just outside the reference range, although the signed median difference was 0.004.

Diagnostic subsets gave the same interpretation. A time-domain subset containing level, spread, change, lag-1 autocorrelation, and run-length entropy had a LOMM discrepancy of 0.1463, within the MAFW reference interval [0.0967, 0.1489]. A local-temporal subset containing change, lag-1 autocorrelation, and run-length entropy also fell within its reference interval. The two robust variance measures together fell within the MAFW reference range. The entropy measures accounted for most of the remaining full-panel discrepancy.

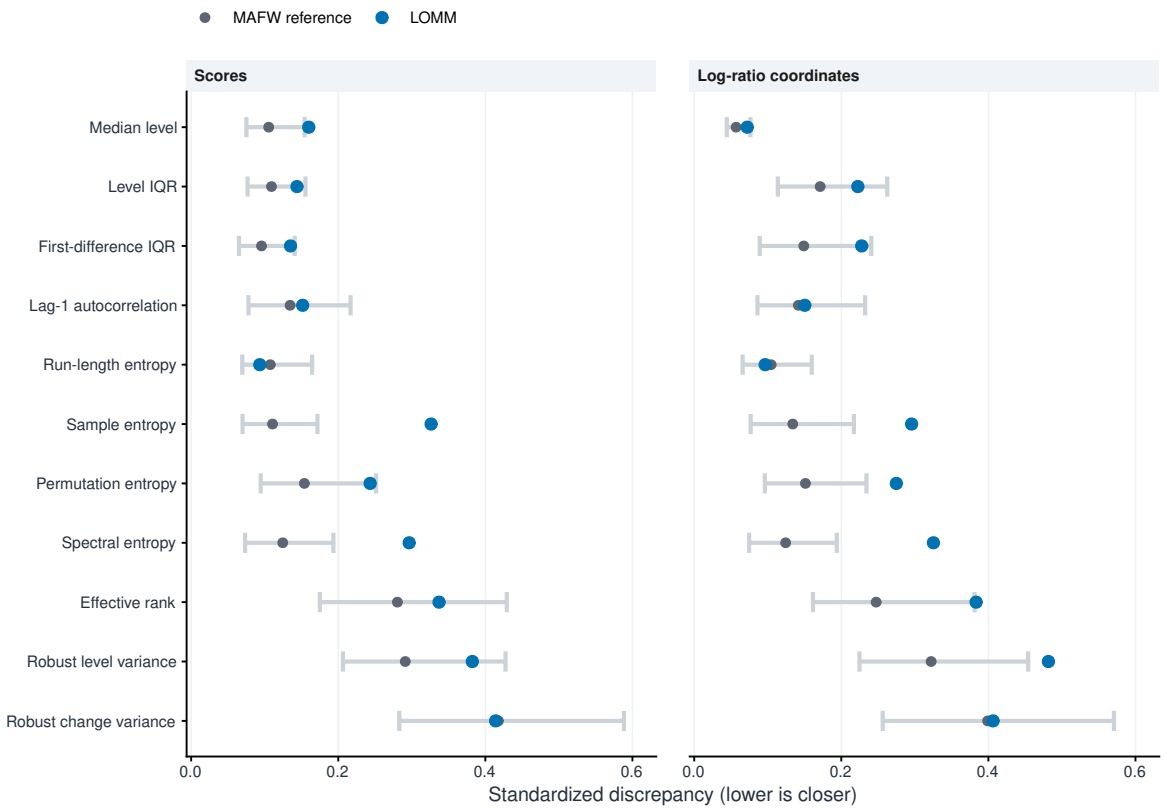



Figure C1: Metric-level discrepancies between LOMM and MAFW classifier-score series. Blue points are medians across 100 LOMM simulations. Gray points and bars show the median and central 95% reference range across 1,000 disjoint MAFW-to-MAFW comparisons. Results are averaged across four folds and shown for scores and isometric log-ratio coordinates. Discrepancy is the larger of the standardized absolute median difference and standardized Wasserstein distance. Smaller values indicate closer agreement.

## C.2 Grouping and Trajectory Diagnostics

This appendix reports two types of post-estimation diagnostic. Grouping diagnostics evaluate recovered indicator groupings among valid network outputs. Trajectory-summary diagnostics compare summaries constructed from recovered groups with the activity-scaled latent trajectories; the trajectories are not direct latent-state estimates produced by the methods.

Table C1: Louvain grouping diagnostics for the matched-method simulation with $T = 100$. Each diagnostic uses the datasets satisfying its stated conditioning rule. Monte Carlo standard errors are computed across simulated datasets, and 95% intervals resample whole datasets, stratified by condition.

| Method | Diagnostic | Estimate | MCSE | 95% interval | Datasets | Valid outputs | Correct-dimensional outputs |
|---|---|---|---|---|---|---|---|
| DynEGA | Signed dimension error | 0.166 | 0.0057 | [0.154, 0.177] | 21600 | 21600 | 20282 |
| DynEGA | Absolute dimension error | 0.168 | 0.0057 | [0.157, 0.180] | 21600 | 21600 | 20282 |
| DynEGA | Adjusted Rand index | 0.935 | 0.0015 | [0.932, 0.938] | 21600 | 21600 | 20282 |
| DynEGA | Variation of information | 0.095 | 0.0023 | [0.091, 0.100] | 21600 | 21600 | 20282 |
| DynEGA | Assignment accuracy given valid network output | 0.965 | 0.0009 | [0.963, 0.966] | 21600 | 21600 | 20282 |
| GraphicalVAR | Signed dimension error | 0.979 | 0.0171 | [0.947, 1.011] | 20329 | 20329 | 16784 |
| GraphicalVAR | Absolute dimension error | 0.979 | 0.0171 | [0.945, 1.012] | 20329 | 20329 | 16784 |
| GraphicalVAR | Adjusted Rand index | 0.867 | 0.0022 | [0.863, 0.872] | 20329 | 20329 | 16784 |
| GraphicalVAR | Variation of information | 0.183 | 0.0031 | [0.177, 0.189] | 20329 | 20329 | 16784 |
| GraphicalVAR | Assignment accuracy given valid network output | 0.902 | 0.0016 | [0.899, 0.906] | 20329 | 20329 | 16784 |
| GIMME | Signed dimension error | 0.319 | 0.0110 | [0.298, 0.341] | 4846 | 4846 | 3795 |
| GIMME | Absolute dimension error | 0.319 | 0.0110 | [0.298, 0.341] | 4846 | 4846 | 3795 |
| GIMME | Adjusted Rand index | 0.733 | 0.0057 | [0.722, 0.744] | 4846 | 4846 | 3795 |
| GIMME | Variation of information | 0.337 | 0.0073 | [0.323, 0.351] | 4846 | 4846 | 3795 |
| GIMME | Assignment accuracy given valid network output | 0.877 | 0.0027 | [0.872, 0.882] | 4846 | 4846 | 3795 |
| Static EGA | Signed dimension error | 0.799 | 0.0159 | [0.770, 0.829] | 21526 | 21526 | 19093 |
| Static EGA | Absolute dimension error | 0.799 | 0.0159 | [0.767, 0.829] | 21526 | 21526 | 19093 |
| Static EGA | Adjusted Rand index | 0.894 | 0.0021 | [0.889, 0.898] | 21526 | 21526 | 19093 |
| Static EGA | Variation of information | 0.147 | 0.0029 | [0.142, 0.153] | 21526 | 21526 | 19093 |
| Static EGA | Assignment accuracy given valid network output | 0.922 | 0.0015 | [0.919, 0.925] | 21526 | 21526 | 19093 |

Table C2: Louvain grouping diagnostics for the matched-method simulation with $T = 25$. Each diagnostic uses the datasets satisfying its stated conditioning rule. Monte Carlo standard errors are computed across simulated datasets, and 95% intervals resample whole datasets, stratified by condition.

| Method | Diagnostic | Estimate | MCSE | 95% interval | Datasets | Valid outputs | Correct-dimensional outputs |
|---|---|---|---|---|---|---|---|
| DynEGA | Signed dimension error | 0.492 | 0.0102 | [0.471, 0.512] | 21600 | 21600 | 18636 |
| DynEGA | Absolute dimension error | 0.494 | 0.0102 | [0.474, 0.515] | 21600 | 21600 | 18636 |
| DynEGA | Adjusted Rand index | 0.902 | 0.0018 | [0.899, 0.906] | 21600 | 21600 | 18636 |
| DynEGA | Variation of information | 0.137 | 0.0025 | [0.132, 0.143] | 21600 | 21600 | 18636 |
| DynEGA | Assignment accuracy given valid network output | 0.935 | 0.0012 | [0.933, 0.938] | 21600 | 21600 | 18636 |
| GraphicalVAR | Signed dimension error | 1.048 | 0.0154 | [1.017, 1.076] | 21579 | 21579 | 16641 |
| GraphicalVAR | Absolute dimension error | 1.048 | 0.0154 | [1.016, 1.080] | 21579 | 21579 | 16641 |
| GraphicalVAR | Adjusted Rand index | 0.859 | 0.0020 | [0.855, 0.863] | 21579 | 21579 | 16641 |
| GraphicalVAR | Variation of information | 0.200 | 0.0028 | [0.195, 0.206] | 21579 | 21579 | 16641 |
| GraphicalVAR | Assignment accuracy given valid network output | 0.893 | 0.0015 | [0.890, 0.896] | 21579 | 21579 | 16641 |
| Static EGA | Signed dimension error | 0.576 | 0.0136 | [0.551, 0.603] | 21541 | 21541 | 19721 |
| Static EGA | Absolute dimension error | 0.576 | 0.0136 | [0.550, 0.603] | 21541 | 21541 | 19721 |
| Static EGA | Adjusted Rand index | 0.922 | 0.0018 | [0.919, 0.926] | 21541 | 21541 | 19721 |
| Static EGA | Variation of information | 0.108 | 0.0025 | [0.103, 0.113] | 21541 | 21541 | 19721 |
| Static EGA | Assignment accuracy given valid network output | 0.943 | 0.0013 | [0.940, 0.945] | 21541 | 21541 | 19721 |

Table C3: Louvain trajectory-summary diagnostics for the matched-method simulation with $T = 100$. Correct-dimensional rows include valid outputs that recovered the correct number of dimensions. All-attempt failure-adjusted rows assign correlation 0 and NRMSE 2 when no valid network output was returned. These are post-estimation summaries of recovered groups; higher correlation and lower NRMSE indicate closer alignment with the activity-scaled latent trajectories.

| Method | Diagnostic | Estimate | MCSE | 95% interval | Datasets | Valid | Correct dimension |
|---|---|---|---|---|---|---|---|
| DynEGA | All-attempt failure-adjusted trajectory correlation | 0.549 | 0.0014 | [0.546, 0.552] | 21600 | 21600 | 20282 |
| DynEGA | All-attempt failure-adjusted trajectory NRMSE | 1.435 | 0.0013 | [1.432, 1.438] | 21600 | 21600 | 20282 |
| GraphicalVAR | All-attempt failure-adjusted trajectory correlation | 0.668 | 0.0020 | [0.664, 0.671] | 21600 | 20329 | 16784 |
| GraphicalVAR | All-attempt failure-adjusted trajectory NRMSE | 0.753 | 0.0036 | [0.746, 0.760] | 21600 | 20329 | 16784 |
| GIMME | All-attempt failure-adjusted trajectory correlation | 0.132 | 0.0019 | [0.129, 0.136] | 21600 | 4846 | 3795 |
| GIMME | All-attempt failure-adjusted trajectory NRMSE | 1.749 | 0.0034 | [1.742, 1.756] | 21600 | 4846 | 3795 |
| Static EGA | All-attempt failure-adjusted trajectory correlation | 0.727 | 0.0017 | [0.723, 0.730] | 21600 | 21526 | 19093 |
| Static EGA | All-attempt failure-adjusted trajectory NRMSE | 0.641 | 0.0031 | [0.635, 0.647] | 21600 | 21526 | 19093 |
| DynEGA | Trajectory correlation (correct-dimensional valid network outputs) | 0.574 | 0.0013 | [0.572, 0.577] | 20282 | 21600 | 20282 |
| DynEGA | Trajectory NRMSE (correct-dimensional valid network outputs) | 1.418 | 0.0014 | [1.416, 1.421] | 20282 | 21600 | 20282 |
| GraphicalVAR | Trajectory correlation (correct-dimensional valid network outputs) | 0.791 | 0.0014 | [0.789, 0.794] | 16784 | 20329 | 16784 |
| GraphicalVAR | Trajectory NRMSE (correct-dimensional valid network outputs) | 0.568 | 0.0031 | [0.562, 0.574] | 16784 | 20329 | 16784 |
| GIMME | Trajectory correlation (correct-dimensional valid network outputs) | 0.606 | 0.0034 | [0.600, 0.613] | 3795 | 4846 | 3795 |
| GIMME | Trajectory NRMSE (correct-dimensional valid network outputs) | 0.898 | 0.0050 | [0.888, 0.907] | 3795 | 4846 | 3795 |
| Static EGA | Trajectory correlation (correct-dimensional valid network outputs) | 0.787 | 0.0014 | [0.784, 0.789] | 19093 | 21526 | 19093 |
| Static EGA | Trajectory NRMSE (correct-dimensional valid network outputs) | 0.563 | 0.0031 | [0.557, 0.569] | 19093 | 21526 | 19093 |

Table C4: Louvain trajectory-summary diagnostics for the matched-method simulation with $T = 25$. Correct-dimensional rows include valid outputs that recovered the correct number of dimensions. All-attempt failure-adjusted rows assign correlation 0 and NRMSE 2 when no valid network output was returned. These are post-estimation summaries of recovered groups; higher correlation and lower NRMSE indicate closer alignment with the activity-scaled latent trajectories.

| Method | Diagnostic | Estimate | MCSE | 95% interval | Datasets | Valid | Correct dimension |
|---|---|---|---|---|---|---|---|
| DynEGA | All-attempt failure-adjusted trajectory correlation | 0.611 | 0.0011 | [0.608, 0.613] | 21600 | 21600 | 18636 |
| DynEGA | All-attempt failure-adjusted trajectory NRMSE | 1.412 | 0.0012 | [1.409, 1.414] | 21600 | 21600 | 18636 |
| GraphicalVAR | All-attempt failure-adjusted trajectory correlation | 0.799 | 0.0012 | [0.796, 0.801] | 21600 | 21579 | 16641 |
| GraphicalVAR | All-attempt failure-adjusted trajectory NRMSE | 0.587 | 0.0023 | [0.582, 0.591] | 21600 | 21579 | 16641 |
| GIMME | All-attempt failure-adjusted trajectory correlation | 0.000 | 0.0000 | [0.000, 0.000] | 21600 | 0 | 0 |
| GIMME | All-attempt failure-adjusted trajectory NRMSE | 2.000 | 0.0000 | [2.000, 2.000] | 21600 | 0 | 0 |
| Static EGA | All-attempt failure-adjusted trajectory correlation | 0.801 | 0.0012 | [0.799, 0.803] | 21600 | 21541 | 19721 |
| Static EGA | All-attempt failure-adjusted trajectory NRMSE | 0.595 | 0.0024 | [0.591, 0.600] | 21600 | 21541 | 19721 |
| DynEGA | Trajectory correlation (correct-dimensional valid network outputs) | 0.654 | 0.0010 | [0.652, 0.655] | 18636 | 21600 | 18636 |
| DynEGA | Trajectory NRMSE (correct-dimensional valid network outputs) | 1.382 | 0.0013 | [1.380, 1.385] | 18636 | 21600 | 18636 |
| GraphicalVAR | Trajectory correlation (correct-dimensional valid network outputs) | 0.871 | 0.0008 | [0.870, 0.873] | 16641 | 21579 | 16641 |
| GraphicalVAR | Trajectory NRMSE (correct-dimensional valid network outputs) | 0.466 | 0.0022 | [0.461, 0.470] | 16641 | 21579 | 16641 |
| Static EGA | Trajectory correlation (correct-dimensional valid network outputs) | 0.836 | 0.0009 | [0.834, 0.838] | 19721 | 21541 | 19721 |
| Static EGA | Trajectory NRMSE (correct-dimensional valid network outputs) | 0.548 | 0.0023 | [0.544, 0.553] | 19721 | 21541 | 19721 |

## C.3 One-at-a-Time Sensitivity Analyses and Scenario Parameters

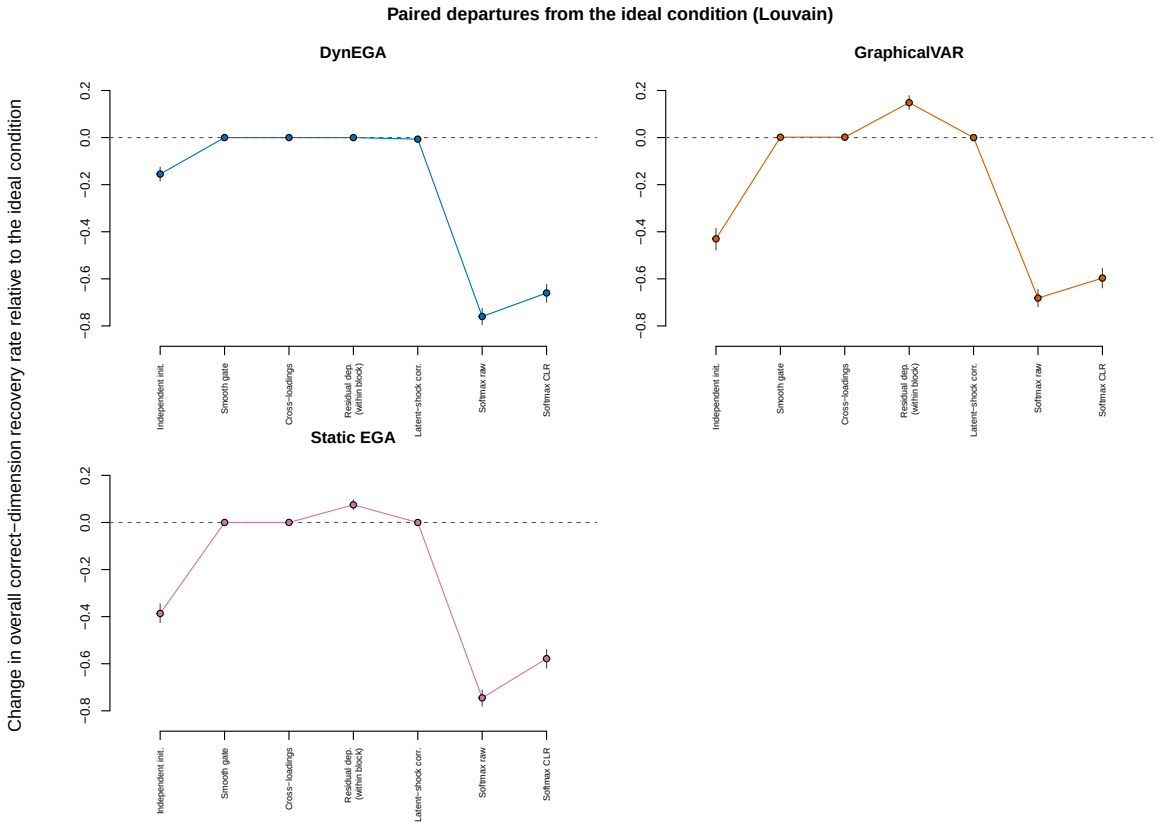



Figure C2: Paired change in overall correct-dimension recovery under each one-at-a-time sensitivity condition. Negative values indicate lower recovery than in the reference condition. Error bars are 95% bootstrap intervals from resampling whole datasets.

The scenario families use the numeric settings below. The displayed $\sigma_q$ vectors are the per-step acceleration-input standard deviations used in the analyses. The primary simulation design with 100 observations uses $\Delta = 0.4$. Conditions with 300 observations occur only in the extended factorial simulation and use $\Delta = 0.25$, so these conditions differ in the number of observations, the interval between them, the total time spanned, and discrete acceleration-input specification.

Codes A–D use asymptotically stable feedback. Codes E–F use $\eta_f < 0$ and $\zeta_f > 0$, which produces oscillatory finite-horizon amplification. Their interpretation is restricted to the stated observation horizons. The simulated-versus-real comparison uses stable negative-$\zeta$ dynamics and is analyzed separately. All oscillatory factors satisfy the cadence restriction $0 < \omega_f < \pi/\Delta$ at the observation intervals used here.

Table C5: Generator-only amplitude diagnostics for the finite-horizon amplification families with $N = 50$, four indicators per dimension, loading 0.60, residual standard deviation 0.25, and 25 replications. The final-tenth energy fraction is the proportion of squared activity-scaled latent signal occurring in the last 10% of observations, averaged across clips within a panel and then summarized across replications.

| Scenario | Dimensions | $T$ | $H$ | Median max $\lvert\tilde{x}\rvert$ | 95th percentile max $\lvert\tilde{x}\rvert$ | Median observed IQR | Final-tenth energy fraction |
|---|---|---|---|---|---|---|---|
| Ranked finite-horizon amplification | 2 | 100 | 39.60 | 55.9 | 61.5 | 3.47 | 0.316 |
| Ranked finite-horizon amplification | 2 | 300 | 74.75 | 329.4 | 376.9 | 7.46 | 0.338 |
| Ranked finite-horizon amplification | 3 | 100 | 39.60 | 62.6 | 73.0 | 1.90 | 0.368 |
| Ranked finite-horizon amplification | 3 | 300 | 74.75 | 430.3 | 478.6 | 4.58 | 0.428 |
| Ranked finite-horizon amplification | 4 | 100 | 39.60 | 85.6 | 102.7 | 1.57 | 0.404 |
| Ranked finite-horizon amplification | 4 | 300 | 74.75 | 994.9 | 1092.7 | 4.02 | 0.498 |
| Switching finite-horizon amplification | 2 | 100 | 39.60 | 69.5 | 81.9 | 4.71 | 0.319 |
| Switching finite-horizon amplification | 2 | 300 | 74.75 | 388.1 | 491.3 | 9.31 | 0.324 |
| Switching finite-horizon amplification | 3 | 100 | 39.60 | 77.7 | 87.5 | 2.33 | 0.367 |
| Switching finite-horizon amplification | 3 | 300 | 74.75 | 518.5 | 608.5 | 5.74 | 0.448 |
| Switching finite-horizon amplification | 4 | 100 | 39.60 | 105.2 | 122.2 | 1.92 | 0.402 |
| Switching finite-horizon amplification | 4 | 300 | 74.75 | 1201.6 | 1350.2 | 4.97 | 0.517 |

Table C6: Activity schedules and dynamic-feedback settings for each scenario family.

| Code | Family | Dynamic family | Switching | Gain | Dwell | Velocity feedback |
|---|---|---|---|---|---|---|
| A | Weak expression, one dominant dimension | Stable template | No | 0.35 / 0.05 | 1.00 / 1.00 | Negative |
| B | Strong expression, one dominant dimension | Stable template | No | 1.00 / 0.08 | 1.00 / 1.00 | Negative |
| C | Strong expression, switching dominant dimension | Stable template | Yes | 1.00 / 0.08 | 0.20 / 0.45 | Negative |
| D | Strong expression, rapid switching | Stable template | Yes | 1.25 / 0.08 | 0.03 / 0.10 | Negative |
| E | Amplifying dynamics, one dominant dimension | Amplification | No | 1.00 / 0.08 | 1.00 / 1.00 | Positive |
| F | Amplifying dynamics, switching dominant dimension | Amplification | Yes | 1.10 / 0.10 | 0.12 / 0.30 | Positive |

Table C7: Dynamic parameters for each scenario family. Acceleration-input SD is the per-step standard deviation $\sigma_q$ used in each condition.

| Code | Dimensions | eta | zeta | Acceleration-input SD |
|---|---|---|---|---|
| A | 2D | -0.30; -0.55 | -1.70; -1.00 | 0.06; 0.15 |
| A | 3D | -0.30; -0.55; -0.85 | -1.70; -1.00; -0.65 | 0.06; 0.15; 0.23 |
| A | 4D | -0.30; -0.55; -0.85; -1.10 | -1.70; -1.00; -0.65; -0.30 | 0.06; 0.15; 0.23; 0.33 |
| B | 2D | -0.30; -0.55 | -1.70; -1.00 | 0.12; 0.30 |
| B | 3D | -0.30; -0.55; -0.85 | -1.70; -1.00; -0.65 | 0.12; 0.30; 0.45 |
| B | 4D | -0.30; -0.55; -0.85; -1.10 | -1.70; -1.00; -0.65; -0.30 | 0.12; 0.30; 0.45; 0.65 |
| C | 2D | -0.30; -0.55 | -1.70; -1.00 | 0.12; 0.30 |
| C | 3D | -0.30; -0.55; -0.85 | -1.70; -1.00; -0.65 | 0.12; 0.30; 0.45 |
| C | 4D | -0.30; -0.55; -0.85; -1.10 | -1.70; -1.00; -0.65; -0.30 | 0.12; 0.30; 0.45; 0.65 |

| | | | | |
|---|---|---|---|---|
| D | 2D | -0.30; -0.55 | -1.70; -1.00 | 0.15; 0.38 |
| D | 3D | -0.30; -0.55; -0.85 | -1.70; -1.00; -0.65 | 0.15; 0.38; 0.56 |
| D | 4D | -0.30; -0.55; -0.85; -1.10 | -1.70; -1.00; -0.65; -0.30 | 0.15; 0.38; 0.56; 0.81 |
| E | 2D | -0.05; -0.10 | 0.08; 0.10 | 0.25; 0.25 |
| E | 3D | -0.05; -0.10; -0.15 | 0.08; 0.10; 0.12 | 0.25; 0.25; 0.25 |
| E | 4D | -0.05; -0.10; -0.15; -0.20 | 0.08; 0.10; 0.12; 0.14 | 0.25; 0.25; 0.25; 0.25 |
| F | 2D | -0.05; -0.10 | 0.08; 0.10 | 0.35; 0.35 |
| F | 3D | -0.05; -0.10; -0.15 | 0.08; 0.10; 0.12 | 0.35; 0.35; 0.35 |
| F | 4D | -0.05; -0.10; -0.15; -0.20 | 0.08; 0.10; 0.12; 0.14 | 0.35; 0.35; 0.35; 0.35 |

Table C8: Deterministic stability diagnostics for the unique dynamic-parameter pairs used across scenario factors. The preceding table maps these pairs to scenarios and factors. The modal exponential-rate factors $g(H) = \exp\{H \max \mathrm{Re}(r)\}$ describe modal rates over the observed horizon rather than total finite-horizon state amplification. $H[100] = 39.6$ and $H[300] = 74.75$ under the stated record designs.

| Scenario codes | eta | zeta | Discriminant | Max Re(root) | rho(Phi[.4]) | rho(Phi[.25]) | g(H[100]) | g(H[300]) | Regime |
|---|---|---|---|---|---|---|---|---|---|
| A, B, C, D | -1.10 | -0.30 | -4.3100 | -0.150 | 0.9418 | 0.9632 | 0.0026 | 0.0000 | Stable oscillatory |
| A, B, C, D | -0.85 | -0.65 | -2.9775 | -0.325 | 0.8781 | 0.9220 | 0.0000 | 0.0000 | Stable oscillatory |
| A, B, C, D | -0.55 | -1.00 | -1.2000 | -0.500 | 0.8187 | 0.8825 | 0.0000 | 0.0000 | Stable oscillatory |
| A, B, C, D | -0.30 | -1.70 | 1.6900 | -0.200 | 0.9231 | 0.9512 | 0.0004 | 0.0000 | Stable nonoscillatory |
| E, F | -0.20 | 0.14 | -0.7804 | 0.070 | 1.0284 | 1.0177 | 15.9906 | 187.2604 | Finite-horizon amplifying oscillatory |
| E, F | -0.15 | 0.12 | -0.5856 | 0.060 | 1.0243 | 1.0151 | 10.7618 | 88.6770 | Finite-horizon amplifying oscillatory |
| E, F | -0.10 | 0.10 | -0.3900 | 0.050 | 1.0202 | 1.0126 | 7.2427 | 41.9929 | Finite-horizon amplifying oscillatory |
| E, F | -0.05 | 0.08 | -0.1936 | 0.040 | 1.0161 | 1.0101 | 4.8744 | 19.8857 | Finite-horizon amplifying oscillatory |

## C.4 Extended Factorial Simulation

The extended factorial simulation evaluated DynEGA on 518,400 datasets. It recovered the correct dimension in 93.7% of those datasets and assigned 95.2% of indicators correctly. Stronger loadings and smaller residual error improved recovery. The design, including the $T = 300$ change in observation interval, is given in Appendix B. Runtime accounting for this design is in Section C.5.

## C.5 Runtime and Feasibility Accounting

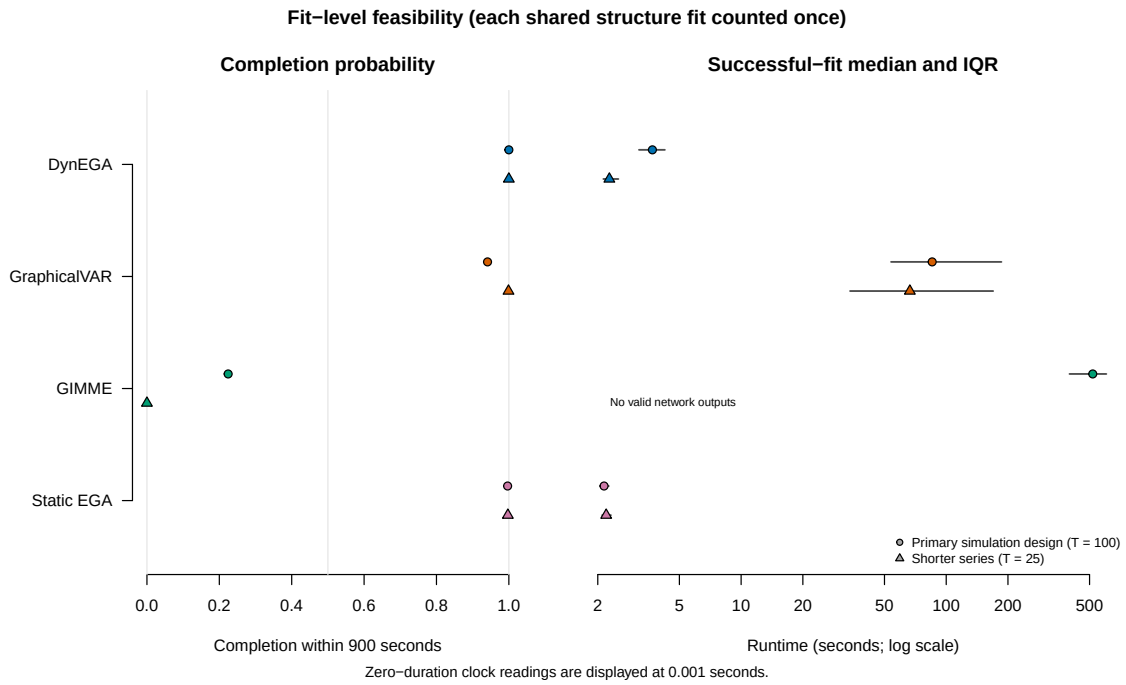



Figure C3: Runtime among valid network outputs in the primary $T = 100$ and shorter $T = 25$ designs. Points show medians and intervals show interquartile ranges on a log10 scale. Counts use unique fitted structures.

Table C9: Runtime and feasibility per method fit in the $T = 100$ primary and $T = 25$ shorter-series matched-method designs. Completion intervals resample whole simulated datasets. Runtime summaries condition on valid network outputs and include structure estimation, network extraction, and community detection. Each fitted structure is reused for Louvain, Leiden, and Walktrap.

| Analysis | Time-series length | Method | Attempts | Valid outputs | Timeouts | Errors | Completion | Runtime median [IQR], s |
|---|---|---|---|---|---|---|---|---|
| Matched-method simulation design | $T = 100$ | DynEGA | 21600 | 21600 | 0 | 0 | 1.000 [1.000, 1.000] | 3.69 [3.18, 4.25] |
| Matched-method simulation design | $T = 100$ | GIMME | 21600 | 4846 | 14477 | 2277 | 0.224 [0.219, 0.230] | 518.16 [398.98, 603.71] |
| Matched-method simulation design | $T = 100$ | GraphicalVAR | 21600 | 20329 | 1271 | 0 | 0.941 [0.938, 0.944] | 85.41 [53.82, 185.91] |
| Matched-method simulation design | $T = 100$ | Static EGA | 21600 | 21526 | 0 | 74 | 0.997 [0.996, 0.997] | 2.15 [2.05, 2.26] |
| Matched-method simulation design | $T = 25$ | DynEGA | 21600 | 21600 | 0 | 0 | 1.000 [1.000, 1.000] | 2.28 [2.13, 2.52] |
| Matched-method simulation design | $T = 25$ | GIMME | 21600 | 0 | 0 | 21600 | 0.000 [0.000, 0.000] | Not available |
| Matched-method simulation design | $T = 25$ | GraphicalVAR | 21600 | 21579 | 21 | 0 | 0.999 [0.999, 0.999] | 66.49 [33.96, 169.39] |
| Matched-method simulation design | $T = 25$ | Static EGA | 21600 | 21541 | 0 | 59 | 0.997 [0.997, 0.998] | 2.20 [2.10, 2.32] |

## C.6 Community-Detection Sensitivity

Louvain, Leiden, and Walktrap were applied to each retained fitted graph. Their results were closely aligned in the matched-method simulation designs (Figure C4) and led to the same interpretation of the one-at-a-time sensitivity conditions (Figure C5). These comparisons reuse fitted structures and therefore assess the community-detection step rather than estimator variability.

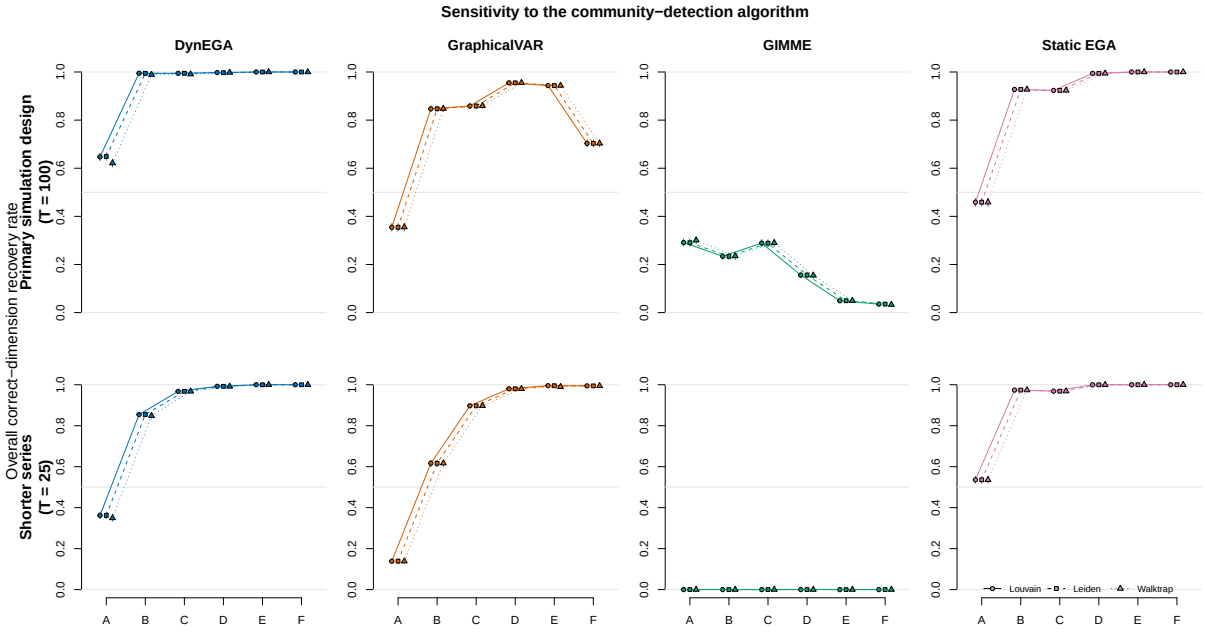



Figure C4: Overall correct-dimension recovery rates for Louvain, Leiden, and Walktrap in the $T = 100$ primary and $T = 25$ shorter-series matched-method designs. Each fitted network is reused across all three algorithms, so differences isolate the community-detection step.

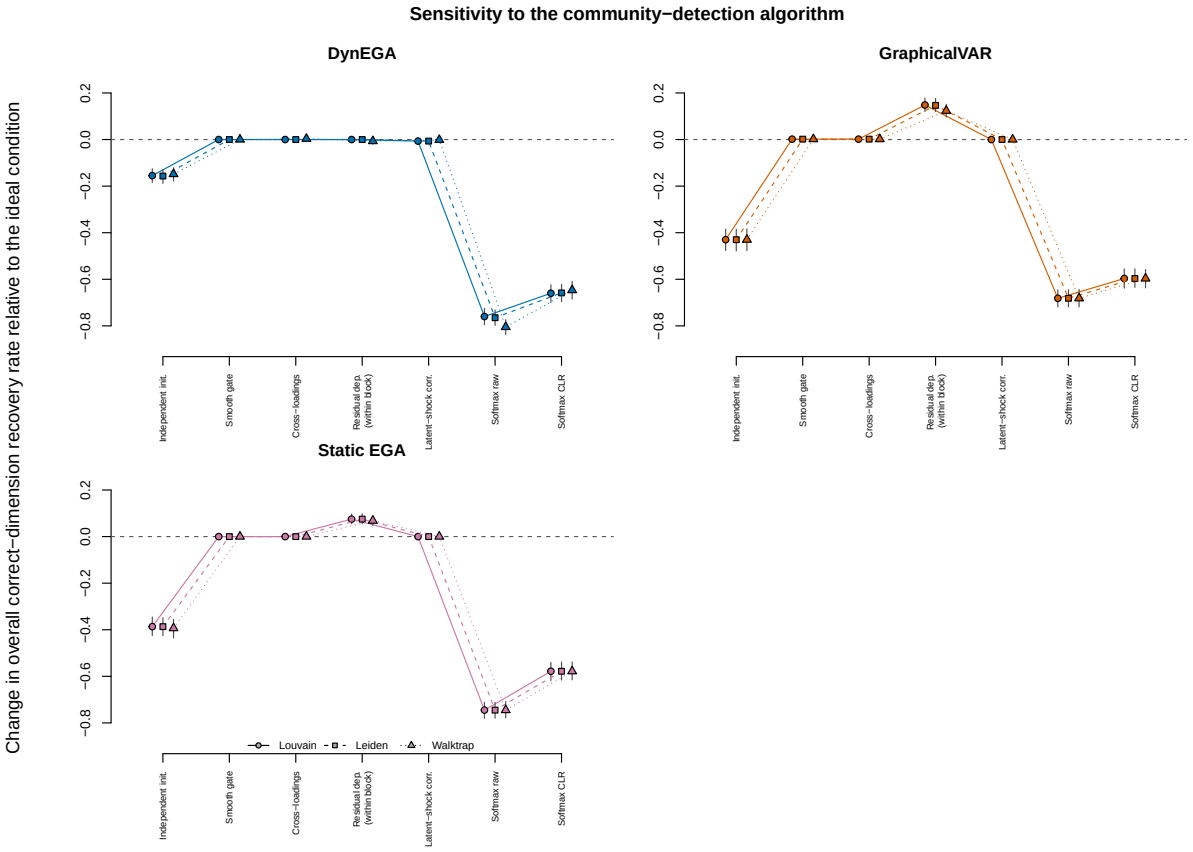



Figure C5: Sensitivity of the paired one-at-a-time effects to Louvain, Leiden, and Walktrap. Values are changes in overall correct-dimension recovery rate relative to the reference condition.